A Textbook for Complex Analysis and Genus One

Complex Tori, Jacobians, and Elliptic Curves

The Three Faces of a Genus-One Group

Complex tori, degree-zero divisor classes, and plane cubics as one genus-one group:

$$\mathbb{C}/\Lambda \;\cong\; \mathrm{Pic}^0(E) \;\cong\; E(\mathbb{C})$$

as complex Lie groups. The chord–tangent law is torus addition transported by

$$z \longmapsto [\wp(z) : \wp'(z) : 1].$$

Spring 2026

Riemann Surfaces and Complex Analysis

Preface

This textbook develops the genus-one bridge between complex analysis, Riemann surface theory, and the algebraic geometry of plane cubics. Its main purpose is to explain why the complex torus, the Jacobian, and the smooth cubic are three presentations of one mathematical object.

The organizing claim is simple to state. A lattice $\Lambda \subset \mathbb{C}$ gives a compact complex Lie group $T = \mathbb{C}/\Lambda$. The Weierstrass functions attach to it a smooth cubic

$$E: \quad Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3,$$

and the map $z \mapsto [\wp(z) : \wp'(z) : 1]$ identifies T with $E(\mathbb{C})$. Divisors identify the same group with $\mathrm{Pic}^0(E)$. Under this identification, the geometric chord–tangent rule on E is just addition on the torus.

The book starts from the exam material: differential forms, Stokes' theorem, residues, compact Riemann surfaces, divisors, and Riemann–Roch. Those topics are not reviewed as a checklist; they are used as the mechanism that makes the main theorem natural.

Big Picture

The goal of this book is to understand one object from three directions. The object is a genus-one complex group. It first appears analytically as a quotient

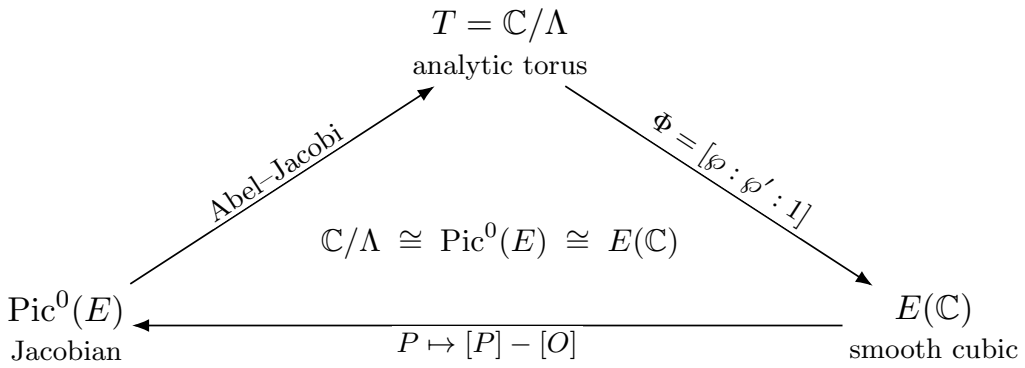
$$T = \mathbb{C}/\Lambda,$$

where addition is inherited from addition in $\mathbb{C}$. It then appears geometrically as a Jacobian,

$$\mathrm{Pic}^0(E),$$

where the group law is encoded by degree-zero divisor classes. Finally it appears algebraically as the complex points of a smooth plane cubic,

$$E(\mathbb{C}): \quad Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3.$$



The central theorem says that these are not merely related examples:

$$\mathbb{C}/\Lambda \cong \mathrm{Pic}^0(E) \cong E(\mathbb{C})$$

as complex Lie groups. This phrase is important: the maps are both biholomorphic and compatible with the group laws. Thus the statement is stronger than an isomorphism of abstract abelian groups; it also preserves the complex structure. The most visible consequence is the chord–tangent law. On the torus, addition is simply

$$[u] + [v] = [u + v].$$

On the cubic, the same addition is disguised as a geometric recipe: draw the line through two points, take the third intersection with the cubic, and reflect it.

What Must Be Understood

1. **Why a torus is a Riemann surface.** A lattice quotient $\mathbb{C}/\Lambda$ is compact, genus one, and still carries a holomorphic group law.

2. **Why elliptic functions are constrained.** Meromorphic periodic functions cannot have arbitrary zeros and poles. The residue theorem and the argument principle force balancing laws.

3. **Why the Weierstrass function creates a cubic.** The functions $\wp$ and $\wp'$ satisfy

$$\wp'^2 = 4\wp^3 - g_2\wp - g_3,$$

   so $z \mapsto [\wp(z) : \wp'(z) : 1]$ sends the torus into a plane cubic.

4. **Why divisors produce the Jacobian.** Principal divisors have degree zero, and Abel's theorem says that on a torus a degree-zero divisor is principal exactly when its Abel sum vanishes.

5. **Why Riemann–Roch gives the correct coordinates.** On a genus-one surface, $\ell(nO) = n$. Thus $1, \wp, \wp'$ are exactly the three functions needed to map the torus into $\mathbb{CP}^2$.

6. **Why line intersection becomes addition.** A line meeting the cubic in P, Q, R cuts out a principal divisor $[P] + [Q] + [R] - 3[O]$. Abel's theorem converts this into the group equation $P + Q + R = O$, so the sum of P and Q is the reflection of R.

The route through the book. The chapters move from the exam tools to the torus, from the torus to elliptic functions, from elliptic functions to the cubic, and from divisors to the Jacobian. The final synthesis is that the analytic, geometric, and algebraic constructions describe the same group.

Notation

$\mathbb{C}, \mathbb{R}, \mathbb{Z}, \mathbb{Q}, \mathbb{F}_p$	complex, real, integer, rational, and finite fields
$\mathbb{H}$	upper half-plane
Λ	lattice $\mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$
$T = \mathbb{C}/\Lambda$	complex torus
O	identity of T, also point at infinity of E
$\wp, \wp'$	Weierstrass function and derivative
g_2, g_3, Δ	invariants and discriminant $g_2^3 - 27g_3^2$
E	cubic $Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3$
$\mathrm{Div}(X)$	divisor group of X
$\mathrm{div}(f)$	principal divisor of f
$\mathrm{Pic}^0(X)$	degree-zero divisor classes modulo principal divisors
$\mathfrak{s}$	Abel–Jacobi sum map
$L(D), \ell(D)$	meromorphic functions bounded by D, and its dimension
Φ	Weierstrass map $[\wp : \wp' : 1]$
$\oplus$	transported group law on E
$\boxplus$	chord–tangent construction
j	j-invariant $1728g_2^3/\Delta$

The symbols in this table are used throughout the book. Later chapters refine some definitions, but this front-matter list fixes the basic notation before the main exposition begins.

Course Guide

Aim of the Course

This twelve-week course gives a route through the book for one two-hour lecture each week. The goal is not to survey elliptic curves broadly, but to understand one central identification in detail:

$$\mathbb{C}/\Lambda \;\cong\; \mathrm{Pic}^0(E) \;\cong\; E(\mathbb{C}).$$

Here the isomorphisms are isomorphisms of complex Lie groups. The course ends by proving that the chord–tangent law on a smooth plane cubic is addition on the torus transported by

$$z \longmapsto [\wp(z) : \wp'(z) : 1].$$

Prerequisites

The course assumes a first course in complex analysis. The following table separates what is used immediately from what the text develops during the course.

Assumed	Holomorphic functions, contour integration, residues, isolated singularities, and basic topology of subsets of $\mathbb{C}$.
Reviewed as needed	Differential forms, Stokes' theorem, meromorphic functions on compact surfaces, and zero-pole counting.
Developed carefully	Compact Riemann surfaces, divisors, Picard groups, Jacobians, genus-one Riemann–Roch, and the Weierstrass embedding.

Structure of the Course

Weeks	Theme	Main objects	Question
1–3	Analytic foundations	Forms, residues, compact surfaces, lattices	How does complex analysis produce a compact genus-one group?
4–6	Elliptic functions	Periodic meromorphic functions, $\wp$, principal divisors	How do functions on $\mathbb{C}/\Lambda$ remember the group law?
7–9	Divisors and Jacobians	$L(D)$, $\mathrm{Pic}^0(E)$, Abel–Jacobi, $[\wp : \wp' : 1]$	Why does the analytic torus become a smooth plane cubic?
10–12	The cubic group law	Lines, tangents, Abel's theorem, j-invariant	Why is chord–tangent addition the same operation as torus addition?

Weekly Lecture Plan

Week	Lecture focus	Reading	Core result
1	Forms, residues, and global obstructions	Ch. 1; App. A	Residues turn local singular data into a global constraint.
2	Compact Riemann surfaces and genus	Ch. 2	Compactness changes complex analysis from local function theory into global geometry.
3	Lattices and complex tori	Ch. 3	$\mathbb{C}/\Lambda$ is a compact genus-one Riemann surface and a complex Lie group.
4	Elliptic functions	Ch. 4	Zeros, poles, and residues of elliptic functions balance on a fundamental parallelogram.
5	The Weierstrass $\wp$-function	Ch. 5	$\wp$ and $\wp'$ satisfy $\wp'^2 = 4\wp^3 - g_2\wp - g_3$.
6	Divisors and principal divisors	Ch. 6	Zeros and poles become degree-zero divisor classes modulo principal divisors.
7	Riemann–Roch in genus one	Ch. 7	$\ell(nO) = n$ for $n \geq 1$, giving the functions needed for the cubic map.
8	Abel's theorem and the Jacobian	Ch. 8	The map $P \mapsto [P] - [O]$ identifies E with $\mathrm{Pic}^0(E)$ as complex Lie groups.
9	The Weierstrass embedding	Ch. 9	$1, \wp, \wp'$ embed the torus into $\mathbb{CP}^2$ as a smooth cubic.
10	The plane cubic group law	Ch. 10	Three collinear points on the cubic have sum O.
11	Transport of addition	Ch. 11; Ch. 14	The chord–tangent operation is torus addition transported by $[\wp : \wp' : 1]$.
12	Synthesis and further directions	Chs. 12–14	Cohomology, moduli, the j-invariant, and finite-field questions extend the same genus-one structure.

Exercises and Written Work

Exercises are meant to force translation between the analytic, divisor-theoretic, and geometric languages. They should be attempted before consulting the solutions appendix.

Work	Main task	Due
Problem Set 1	Residues, compactness, genus, and quotient tori	After Week 3
Problem Set 2	Elliptic functions, $\wp$, and principal divisors	After Week 6
Problem Set 3	Riemann–Roch, Abel–Jacobi, and the Weierstrass embedding	After Week 9
Problem Set 4	Line divisors, tangents, and transported addition	After Week 11
Final synthesis	A concise proof narrative of $\mathbb{C}/\Lambda \cong \mathrm{Pic}^0(E) \cong E(\mathbb{C})$ as complex Lie groups	Week 12

Final target. By the end of the course, the student should be able to read the equation of the Weierstrass cubic, the divisor class $[P] - [O]$, and the point $z \in \mathbb{C}/\Lambda$ as three notations for the same genus-one group element.

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Chapter 1

Exam Background: Forms, Residues, and Global Obstructions

The main theorem concerns a torus, a Jacobian, and a cubic. The proof begins much earlier, with a lesson from the midterm: a local calculation can fail globally because a differential form carries an obstruction. The winding form on $\mathbb{R}^2 \setminus \{0\}$,

$$\omega_{\text{wind}} = \frac{-y\,dx + x\,dy}{x^2 + y^2},$$

is closed, but $\int_{S^1} \omega_{\text{wind}} = 2\pi$. It is locally $d\theta$, but no global single-valued angle function exists.

Definition 1.1. An obstruction is a locally invisible quantity that is detected by integration around a global cycle. In complex analysis the model obstruction is the residue:

$$\frac{1}{2\pi i} \int_\gamma \frac{f'(z)}{f(z)}\,dz.$$

The same idea drives the later theory of divisors. Zeros and poles of a meromorphic function are local data. On a compact surface they cannot be chosen freely, because residues and periods force global balancing laws.

Proposition 1.2 (The exam mechanism)**.** *The sequence*

$$\textit{forms} \longrightarrow \textit{Stokes} \longrightarrow \textit{residues} \longrightarrow \textit{global constraints}$$

is the mechanism behind the divisor statements used later.

Proof. Stokes' theorem turns the integral of an exact form over a boundary into an interior integral. If a meromorphic form is regular except at finitely many points, deleting small disks around those points makes Stokes applicable. Letting the disk radii shrink leaves only small circle integrals, hence residues. On a compact surface the outer boundary is absent, so the sum of all local residues must vanish. □

Why this matters. The statement that principal divisors have degree zero is the divisor version of the residue theorem applied to df/f. Abel's theorem is a refined version: not only the total number of zeros and poles but also their Abel sum is constrained.

Exercises

Intent. Check, by direct differentiation, that the local winding form is closed away from its singularity.

Exercise 1.1. Write $\omega_{\text{wind}} = P\,dx + Q\,dy$. Compute Q_x, compute P_y, and subtract them to verify directly that

$$d\omega_{\text{wind}} = (Q_x - P_y)\,dx \wedge dy = 0$$

on $\mathbb{R}^2 \setminus \{0\}$.

Intent. Compute the period of the winding form on the basic loop and see the obstruction to exactness.

Exercise 1.2. Parametrize the unit circle by $x = \cos t$, $y = \sin t$. Compute dx, dy, the numerator $-y\,dx + x\,dy$, and then the integral $\int_{S^1} \omega_{\text{wind}}$.

Intent. Relate the local order of a meromorphic function to the residue of its logarithmic derivative.

Exercise 1.3. Let f be meromorphic near p. Write $f = (z-p)^m u(z)$ with $u(p) \neq 0$. Compute df/f explicitly and read off the coefficient of $(z-p)^{-1}\,dz$, showing that it is $\operatorname{ord}_p(f)$.

Intent. Turn the compact residue theorem into the degree-zero property of principal divisors.

Exercise 1.4. Assume X is compact and f is a nonzero meromorphic function on X. Compute

$$\sum_p \operatorname{Res}_p(df/f)$$

in two ways: first by the compact residue theorem, and second using the previous exercise. Conclude that $\deg \operatorname{div}(f) = 0$.

Chapter 2

Compact Riemann Surfaces and Genus

A Riemann surface is the natural setting for one-variable complex analysis after we stop insisting that all functions live on open subsets of $\mathbb{C}$. It is a topological surface with charts to $\mathbb{C}$ whose transition functions are holomorphic. Compact Riemann surfaces are the complex-analytic analogues of closed algebraic curves.

Definition 2.1. A compact Riemann surface X has genus g if its underlying oriented topological surface has g handles. Equivalently, $\chi(X) = 2 - 2g$.

The midterm computation that the real torus has Euler characteristic 0 is therefore already telling us that a complex torus has genus 1. This number is not cosmetic. It is exactly the number that makes Riemann–Roch read $\ell(D) = \deg D$ for positive divisors on an elliptic curve.

Proposition 2.2. *Every holomorphic function on a compact connected Riemann surface is constant.*

Proof. The image of a compact surface under a holomorphic function is compact in $\mathbb{C}$, so the function has a maximum absolute value. The maximum modulus principle then forces the function to be locally constant, hence constant. □

Definition 2.3. A meromorphic function on a Riemann surface is a holomorphic map to $\mathbb{CP}^1$. Its poles are the points mapping to ∞.

Compactness changes the rules. On a compact surface, nonconstant holomorphic functions disappear, but meromorphic functions become structured maps to $\mathbb{CP}^1$. The map $\wp : T \to \mathbb{CP}^1$ is exactly such a map.

Exercises

Intent. Convert Euler characteristic into genus in the genus-one case.

Exercise 2.1. Substitute $\chi = 0$ into $\chi = 2 - 2g$ and solve the resulting linear equation for g.

Intent. Identify exactly where compactness enters the maximum modulus argument.

Exercise 2.2. Use the concrete function $f(z) = z$ on $\mathbb{C}$. Compute why $|f|$ has no maximum on $\mathbb{C}$, and use this calculation to identify the step in the maximum modulus proof that fails on noncompact surfaces.

Intent. Use compactness and local normal forms to prove zeros and poles are finite.

Exercise 2.3. Suppose a meromorphic function on a compact surface had infinitely many zeros or poles. Use compactness to produce an accumulation point, then use the local forms $(z-p)^m u(z)$ and $(z-p)^{-m} u(z)$ to compute why this contradicts isolation of zeros and poles.

Intent. Recover the torus genus by a concrete cell count on the glued parallelogram.

Exercise 2.4. The quotient of a parallelogram by identifying opposite sides is a topological torus. Count the cell classes F, E, V after the identifications, compute

$$\chi = F - E + V,$$

and then use $\chi = 2 - 2g$ to compute the genus.

Chapter 3

Lattices and Complex Tori

Let $\omega_1, \omega_2 \in \mathbb{C}$ be real-linearly independent. The lattice

$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$$

tiles the complex plane by translates of a parallelogram. The quotient

$$T = \mathbb{C}/\Lambda$$

is obtained by identifying points that differ by lattice elements.

Definition 3.1. The complex torus associated to Λ is the quotient group $T = \mathbb{C}/\Lambda$, equipped with the complex structure induced from small disks in $\mathbb{C}$.

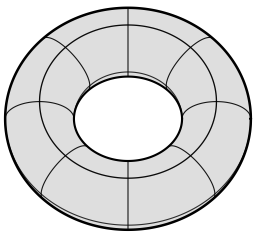
Addition in $\mathbb{C}$ descends to addition on T:

$$[u] + [v] = [u + v].$$

This is the analytic group law. It is associative, commutative, and has identity $[0]$ because those facts were true before quotienting.

Proposition 3.2. *The quotient $T = \mathbb{C}/\Lambda$ is a compact genus-one Riemann surface and a complex Lie group.*

Proof. A fundamental parallelogram is compact, and its quotient maps onto T, so T is compact. Away from identified boundary points, charts come directly from the plane; near a boundary identification one takes a small disk whose translates are disjoint and uses its projection. The quotient of a parallelogram by opposite-side identifications is a topological torus, hence genus one. Addition is holomorphic because it is induced by the holomorphic addition map on $\mathbb{C}$. □



A real picture of the quotient surface.
The complex structure comes from the plane before gluing.

Exercises

Intent. Verify that addition on the quotient is independent of chosen representatives.

Exercise 3.1. Let $u' = u + \lambda$ and $v' = v + \mu$ with $\lambda, \mu \in \Lambda$. Compute $u' + v'$ and show explicitly that it differs from $u + v$ by an element of Λ. Conclude that addition on $\mathbb{C}/\Lambda$ is well-defined.

Intent. Check that an integral change of basis preserves the same lattice.

Exercise 3.2. Let

$$\begin{pmatrix} \omega_1' \\ \omega_2' \end{pmatrix} = A \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix}, \qquad A \in \mathrm{GL}_2(\mathbb{Z}).$$

Use the integer entries of A and A^{-1} to prove by direct inclusion that $\mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 = \mathbb{Z}\omega_1' + \mathbb{Z}\omega_2'$.

Intent. Normalize a lattice basis so its period ratio lies in the upper half-plane.

Exercise 3.3. Let $\tau = \omega_2/\omega_1$. First compute why $\operatorname{Im}\tau \neq 0$. If $\operatorname{Im}\tau < 0$, swap the basis vectors and compute $\operatorname{Im}(1/\tau)$. Conclude that an oriented basis can be chosen with $\tau \in \mathbb{H}$.

Intent. Check that torus translations are well-defined biholomorphic group translations.

Exercise 3.4. Let $a \in \mathbb{C}$. Show that the translation map

$$T \longrightarrow T, \qquad [z] \longmapsto [z + a]$$

is a well-defined biholomorphic group translation by computing what happens when z is replaced by $z + \lambda$, and by writing down the inverse translation.

Intent. Separate complex Lie group isomorphism from a mere abstract group isomorphism.

Exercise 3.5. In the final theorem, the symbol $\cong$ means isomorphism of complex Lie groups. Make a three-row checklist: group homomorphism, holomorphic map, and holomorphic inverse. For each row, write the equation or property that must be verified beyond an abstract abelian-group isomorphism.

Chapter 4

Elliptic Functions on $\mathbb{C}/\Lambda$

A meromorphic function on $T = \mathbb{C}/\Lambda$ is the same thing as a meromorphic function f on $\mathbb{C}$ satisfying $f(z + \lambda) = f(z)$ for every $\lambda \in \Lambda$. Such functions are called elliptic functions.

Definition 4.1. An elliptic function for Λ is a meromorphic function $f : \mathbb{C} \to \mathbb{CP}^1$ that is Λ-periodic.

Compactness immediately rules out holomorphic examples.

Proposition 4.2. *A holomorphic elliptic function is constant.*

Proof. It descends to a holomorphic function on the compact surface T, so proposition 2.2 applies. □

The useful elliptic functions must therefore have poles. The exam residue method now gives three constraints. In one period parallelogram, an elliptic function has residue sum zero, as many zeros as poles counted with multiplicity, and the weighted sum of its zeros minus the weighted sum of its poles lies in Λ.

Theorem 4.3 (Elliptic Liouville laws)**.** *Let f be a nonzero elliptic function. In a fundamental parallelogram avoiding zeros and poles on its boundary,*

$$\sum \operatorname{Res}(f\,dz) = 0, \qquad \sum \operatorname{ord}_a(f) = 0, \qquad \sum_{\operatorname{ord}_a(f)>0} \operatorname{ord}_a(f)a - \sum_{\operatorname{ord}_b(f)<0} (-\operatorname{ord}_b(f))b \in \Lambda.$$

Proof sketch. Integrate $f\,dz$, $f'/f\,dz$, and $zf'/f\,dz$ around the boundary of a period parallelogram. Periodicity makes opposite sides cancel in the first two integrals. In the third integral, opposite sides cancel up to periods, which is why the final answer is only well-defined modulo Λ. □

The bridge to Abel. The third law is already Abel's theorem in its torus form: a principal divisor of degree zero has Abel sum zero in $\mathbb{C}/\Lambda$.

Exercises

Intent. Use the residue obstruction to rule out a single simple pole on a torus.

Exercise 4.1. Assume a nonconstant elliptic function has total pole order 1. Write the Laurent expansion at its unique simple pole, compute the residue contribution, and use the residue-sum law to obtain a contradiction. Conclude that the total pole order is at least 2.

Intent. Compute boundary cancellation on a period parallelogram via the argument principle.

Exercise 4.2. Apply the argument principle on a fundamental parallelogram P:

$$\frac{1}{2\pi i}\int_{\partial P}\frac{f'}{f}\,dz.$$

Pair the four side integrals using periodicity and compute why the total is 0. Conclude that the number of zeros equals the number of poles, counted with multiplicity.

Intent. See why Abel sums naturally live modulo the lattice.

Exercise 4.3. Let a lift $a_i \in \mathbb{C}$ of a point of T be replaced by $a_i + \lambda_i$, with $\lambda_i \in \Lambda$. Compute how $\sum_i m_i a_i$ changes, and conclude that the Abel sum is naturally a point of $\mathbb{C}/\Lambda$, not of $\mathbb{C}$.

Intent. Verify the descent/ascent between elliptic functions and meromorphic functions on the quotient.

Exercise 4.4. Let $q : \mathbb{C} \to T = \mathbb{C}/\Lambda$ be the quotient map. Show that a meromorphic function F on T pulls back to an elliptic function by computing $(F \circ q)(z + \lambda)$. Conversely, if $f(z + \lambda) = f(z)$, compute why the formula $F([z]) = f(z)$ is independent of the chosen representative.

Intent. Compute an explicit principal-divisor Abel sum and connect it to $\mathfrak{s}(\mathrm{div}(f)) = 0$.

Exercise 4.5. Let f be a nonzero elliptic function with divisor

$$\mathrm{div}(f) = [a_1] + [a_2] - [b_1] - [b_2]$$

on T, with chosen lifts $a_1, a_2, b_1, b_2 \in \mathbb{C}$. Use the third identity in theorem 4.3 to compute

$$a_1 + a_2 - b_1 - b_2 \pmod{\Lambda}.$$

Then rewrite the calculation as $\mathfrak{s}(\mathrm{div}(f)) = 0$.

Chapter 5

The Weierstrass $\wp$-Function

The analytic theory of elliptic curves centers on the Weierstrass function

$$\wp(z) = \frac{1}{z^2} + \sum_{\lambda \in \Lambda \setminus \{0\}} \left(\frac{1}{(z-\lambda)^2} - \frac{1}{\lambda^2} \right).$$

The subtraction term makes the series converge normally on compact sets away from Λ. The function is even, Λ-periodic, and has one double pole per fundamental parallelogram.

Definition 5.1. The Eisenstein series of weight k is

$$G_k(\Lambda) = \sum_{\lambda \in \Lambda \setminus \{0\}} \lambda^{-k}$$

for $k > 2$. Set $g_2 = 60G_4$ and $g_3 = 140G_6$.

Theorem 5.2 (The differential equation)**.** *The Weierstrass function satisfies*

$$\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3.$$

Proof sketch. Compare Laurent expansions at 0. Both sides have principal part $4z^{-6}$, and the constants g_2, g_3 are chosen so that the remaining negative and constant terms cancel. The difference is then a holomorphic elliptic function, hence constant by proposition 4.2; its Laurent expansion shows that constant is 0. □

Corollary 5.3. *The functions $x = \wp(z)$ and $y = \wp'(z)$ land on the affine cubic*

$$y^2 = 4x^3 - g_2x - g_3.$$

This is the exact moment where the analytic face becomes algebraic. The final exam's minimal-polynomial problem for $\wp'$ over $\mathbb{C}(\wp)$ is the algebraic shadow of this differential equation.

Exercises

Intent. Prove evenness of $\wp$ by reindexing the defining lattice sum.

Exercise 5.1. Use the defining lattice sum for $\wp$. Substitute $-z$ for z, then reindex the lattice terms by $\lambda \mapsto -\lambda$ to show by calculation that $\wp(-z) = \wp(z)$.

Intent. Use the chain rule to derive the oddness of $\wp'$.

Exercise 5.2. Differentiate the identity $\wp(-z) = \wp(z)$ with respect to z. Keep track of the chain-rule minus sign and compute the resulting parity of $\wp'$.

Intent. Read the Weierstrass differential equation as algebraicity over $\mathbb{C}(\wp)$.

Exercise 5.3. Assuming theorem 5.2, write an explicit quadratic polynomial

$$Y^2 - (4\wp^3 - g_2\wp - g_3) \in \mathbb{C}(\wp)[Y]$$

that is satisfied by $Y = \wp'$. Use this calculation to show that $\wp'$ is algebraic of degree at most 2 over $\mathbb{C}(\wp)$.

Intent. Extract the pole orders that will feed the genus-one Riemann–Roch calculation.

Exercise 5.4. Using the Laurent expansions

$$\wp(z) = z^{-2} + O(z^2), \qquad \wp'(z) = -2z^{-3} + O(z),$$

compute the pole orders of $\wp$ and $\wp'$ at $O \in T$.

Intent. Substitute the Weierstrass functions into the cubic equation.

Exercise 5.5. Set $x = \wp(z)$ and $y = \wp'(z)$. Substitute these expressions into theorem 5.2 and verify directly that the pair (x, y) lies on the affine cubic

$$y^2 = 4x^3 - g_2x - g_3.$$

Chapter 6

Divisors and Principal Divisors

A divisor records zeros and poles as a formal integer combination of points.

Definition 6.1. For a compact Riemann surface X, a divisor is a finite sum

$$D = \sum n_p[p], \qquad n_p \in \mathbb{Z}.$$

Its degree is $\deg D = \sum n_p$. If f is meromorphic, its principal divisor is

$$\operatorname{div}(f) = \sum_p \operatorname{ord}_p(f)[p].$$

Principal divisors are the geometric version of logarithmic derivatives. Since $\operatorname{Res}_p(df/f) = \operatorname{ord}_p(f)$, the compact residue theorem gives:

Proposition 6.2. *Every principal divisor on a compact Riemann surface has degree zero.*

Proof. The meromorphic one-form df/f has residue $\operatorname{ord}_p(f)$ at p. The sum of residues on a compact Riemann surface is zero, so $\deg \operatorname{div}(f) = 0$. □

Definition 6.3. The degree-zero divisor group is $\operatorname{Div}^0(X)$. The principal divisors form a subgroup $\operatorname{Prin}(X) \subset \operatorname{Div}^0(X)$, and

$$\operatorname{Pic}^0(X) = \operatorname{Div}^0(X)/\operatorname{Prin}(X).$$

The group $\operatorname{Pic}^0(E)$ is not decorative notation. It is the geometric home of the same group already seen analytically as $\mathbb{C}/\Lambda$. A point $[P] - [O]$ is a degree-zero divisor class, and the map $P \mapsto [P] - [O]$ will identify the curve with its Jacobian.

Exercises

Intent. Practice divisor bookkeeping, including the contribution at infinity.

Exercise 6.1. Compute the divisor of $z^2(z-1)^{-3}$ on $\mathbb{CP}^1$. First compute the orders at 0 and 1. Then use the coordinate $w = 1/z$ to compute the order at ∞.

Intent. Verify that principal divisors form a degree-zero subgroup.

Exercise 6.2. For meromorphic functions f, g, compute $\operatorname{div}(fg)$, $\operatorname{div}(f^{-1})$, and $\operatorname{div}(c)$ for $c \in \mathbb{C}^*$. Use these calculations, together with degree zero for principal divisors, to show that $\operatorname{Prin}(X)$ is a subgroup of $\operatorname{Div}^0(X)$.

Intent. Check the basic degree-zero divisor attached to a pointed curve.

Exercise 6.3. Compute the degree of $[P] - [O]$ by adding its two coefficients.

Intent. Compute addition of degree-zero divisor classes in Pic^0.

Exercise 6.4. Show that

$$([P] - [O]) + ([Q] - [O]) = [P] + [Q] - 2[O]$$

defines an element of $\mathrm{Pic}^0(X)$ by explicitly computing the degree of the right-hand side and then passing modulo $\mathrm{Prin}(X)$.

Intent. Translate equality in Pic^0 into a principal-divisor condition.

Exercise 6.5. Let $D, D' \in \mathrm{Div}^0(X)$. Compute the difference $D - D'$ in the quotient $\mathrm{Div}^0(X)/\mathrm{Prin}(X)$. State and prove the criterion for D and D' to define the same element of $\mathrm{Pic}^0(X)$.

Chapter 7

Riemann–Roch in Genus One

For a divisor D, define

$$L(D) = \{f \in \mathcal{M}(X) : \operatorname{div}(f) + D \geq 0\} \cup \{0\}.$$

Thus $L(D)$ is the vector space of meromorphic functions whose poles are bounded by D.

Theorem 7.1 (Riemann–Roch)**.** *On a compact Riemann surface of genus g,*

$$\ell(D) - \ell(K - D) = \deg D + 1 - g,$$

where K is a canonical divisor and $\ell(D) = \dim_{\mathbb{C}} L(D)$.

We will use the theorem in its genus-one form. On a complex torus, dz is a nowhere-vanishing holomorphic 1-form, so $K \sim 0$. Since $g = 1$, the formula becomes

$$\ell(D) - \ell(-D) = \deg D.$$

If $\deg D > 0$, then $L(-D) = 0$, and therefore $\ell(D) = \deg D$.

Corollary 7.2. *For $T = \mathbb{C}/\Lambda$ and $O = [0]$,*

$$\ell(nO) = n \qquad (n \geq 1).$$

This is the dimension count behind the fact that $1, \wp, \wp'$ span $L(3O)$. The function 1 has no pole, $\wp$ has a double pole at O, and $\wp'$ has a triple pole at O. Since $\ell(3O) = 3$, these three functions exhaust the space.

Warning 7.3. Riemann–Roch is a deep theorem in general. In this book it is used as a standard final-exam tool and interpreted carefully in genus one. Claims that require more algebraic geometry are explicitly labeled as such.

Exercises

Intent. Use genus-one Riemann–Roch as a dimension calculator.

Exercise 7.1. Using $\ell(nO) = n$, compute $\dim L(O)$, $\dim L(2O)$, and $\dim L(3O)$ by substituting $n = 1, 2, 3$.

Intent. Show why $L(O)$ contains no nonconstant function.

Exercise 7.2. Split into two cases. If $f \in L(O)$ has no pole, use compactness to compute that it is constant. If f has a simple pole at O, compute its total pole order and compare with the elliptic-function pole-order obstruction from Chapter 4.

Intent. Use pole orders to prove linear independence of $1, \wp, \wp'$.

Exercise 7.3. Suppose $a + b\wp + c\wp' = 0$. Compare the pole orders $0, 2, 3$ at O and use the largest pole order term to compute successively that $c = b = a = 0$.

Intent. Combine membership, pole-order independence, and dimension to identify $L(2O)$.

Exercise 7.4. Use $\ell(2O) = 2$ and the pole order of $\wp$ to prove

$$L(2O) = \operatorname{span}_{\mathbb{C}}(1, \wp).$$

Your proof should explicitly check membership of $1, \wp$, linear independence by pole order, and the dimension count.

Intent. Identify $L(3O)$, the coordinate space behind the Weierstrass map.

Exercise 7.5. Use $\ell(3O) = 3$ and the pole orders of $\wp$ and $\wp'$ to prove

$$L(3O) = \operatorname{span}_{\mathbb{C}}(1, \wp, \wp').$$

Your proof should explicitly check membership, pole-order independence, and the dimension count.

Chapter 8

Abel's Theorem and the Jacobian

Let $T = \mathbb{C}/\Lambda$. If

$$D = \sum n_i[p_i] \in \mathrm{Div}^0(T),$$

choose lifts $z_i \in \mathbb{C}$ of p_i and define

$$\mathfrak{s}(D) = \sum n_i z_i \pmod{\Lambda}.$$

Because $\sum n_i = 0$, changing lifts changes the sum by an element of Λ. Thus $\mathfrak{s}(D) \in T$ is well-defined.

Theorem 8.1 (Abel's theorem on a torus)**.** *A degree-zero divisor D on T is principal if and only if $\mathfrak{s}(D) = 0$ in T.*

Proof sketch. If $D = \mathrm{div}(f)$, the elliptic Liouville laws of theorem 4.3 show that the Abel sum vanishes. Conversely, if the degree and Abel sum vanish, one constructs an elliptic function with exactly the prescribed zeros and poles. A standard construction uses the Weierstrass σ-function; Riemann–Roch gives another route. □

Corollary 8.2. *The Abel–Jacobi map induces an isomorphism*

$$\mathrm{Pic}^0(T) \cong T = \mathbb{C}/\Lambda.$$

Transporting this statement to the cubic gives the geometric identification

$$\mathrm{Pic}^0(E) \cong E(\mathbb{C}).$$

The map is $P \mapsto [P] - [O]$, and Abel's theorem tells us exactly when two such degree-zero divisor classes are equal.

Exercises

Intent. Compute the Abel sum of the simplest degree-zero divisor.

Exercise 8.1. Choose lifts $p, q \in \mathbb{C}$ of $P, Q \in T$. Compute $\mathfrak{s}([P] - [Q])$ from the definition and show that it is the class of $p - q$ modulo Λ, hence $P - Q$ in the group T.

Intent. Use Abel's theorem to recognize when a two-point divisor is principal.

Exercise 8.2. Choose lifts $p, q, 0 \in \mathbb{C}$ of P, Q, O. Compute the Abel sum of $[P] + [Q] - 2[O]$. Use Abel's theorem to turn the congruence $p + q \equiv 0 \pmod{\Lambda}$ into a principal-divisor criterion.

Intent. Use Abel sums to prove injectivity of the point-to-divisor-class map.

Exercise 8.3. Assume $[P] - [O] = [Q] - [O]$ in $\mathrm{Pic}^0(T)$. Subtract the two divisor classes, compute the Abel sum of $[P] - [Q]$, and conclude that $P = Q$.

Intent. Check additivity of the Abel-sum map term by term.

Exercise 8.4. Write

$$D = \sum_i n_i[p_i], \qquad D' = \sum_j m_j[q_j],$$

with chosen lifts of all points. Compute both sides of

$$\mathfrak{s}(D + D') = \mathfrak{s}(D) + \mathfrak{s}(D')$$

term by term in $\mathbb{C}/\Lambda$.

Intent. Verify that principal divisors are killed so the Abel-sum map descends to Pic^0.

Exercise 8.5. Let $D \in \mathrm{Div}^0(T)$ and let f be a nonzero meromorphic function on T. Compute

$$\mathfrak{s}(D + \mathrm{div}(f)) - \mathfrak{s}(D).$$

Use Abel's theorem to show that this difference is 0, and then verify that the induced map $\mathrm{Pic}^0(T) \to T$ is a group homomorphism.

Chapter 9

The Weierstrass Embedding into $\mathbb{CP}^2$

The three functions spanning $L(3O)$ define a projective map

$$\Phi : T \longrightarrow \mathbb{CP}^2, \qquad z \longmapsto [\wp(z) : \wp'(z) : 1],$$

with O sent to $[0 : 1 : 0]$. By theorem 5.2, its image lies on

$$E : \quad Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3.$$

Theorem 9.1. *If $\Delta = g_2^3 - 27g_3^2 \neq 0$, the map Φ is a biholomorphism from $\mathbb{C}/\Lambda$ onto the smooth projective cubic E.*

Proof sketch. Near O, the Laurent expansions $\wp(z) = z^{-2} + \cdots$ and $\wp'(z) = -2z^{-3} + \cdots$ show that the map extends holomorphically to $[0 : 1 : 0]$. The equation shows the image lies on E. Smoothness follows from $\Delta \neq 0$. The evenness of $\wp$ and oddness of $\wp'$ distinguish the two points over a generic x-value, and the special half-period points are accounted for by ramification. Compactness then promotes the holomorphic bijection to a biholomorphism. □

The analytic-to-algebraic map. The map $\Phi = [\wp : \wp' : 1]$ is not only a parametrization. It is the identification that allows the elementary group law on $\mathbb{C}/\Lambda$ to become a geometric law on a cubic.

Exercises

Intent. Compute the projective limit that sends the lattice origin to the point at infinity.

Exercise 9.1. Use

$$\wp(z) = z^{-2} + O(z^2), \qquad \wp'(z) = -2z^{-3} + O(z)$$

and multiply the homogeneous coordinates $[\wp(z) : \wp'(z) : 1]$ by z^3. Compute the limit as $z \to 0$ and obtain $\Phi(0) = [0 : 1 : 0]$.

Intent. Track how $z \mapsto -z$ changes the affine Weierstrass coordinates.

Exercise 9.2. Using $\wp(-z) = \wp(z)$ and $\wp'(-z) = -\wp'(z)$, compute

$$x(-z), \qquad y(-z)$$

for $x(z) = \wp(z)$, $y(z) = \wp'(z)$.

Intent. Detect singularity of the affine cubic by solving the partial-derivative equations.

Exercise 9.3. For

$$F(x, y) = y^2 - 4x^3 + g_2x + g_3,$$

compute F_x and F_y. Solve the equations

$$F = F_x = F_y = 0$$

and show that they are equivalent to $y = 0$ and x being a repeated root of $4x^3 - g_2x - g_3$.

Intent. Convert parity of $\wp, \wp'$ into cubic reflection.

Exercise 9.4. Compute the homogeneous coordinates of $\Phi(-z)$ and use the parity of $\wp$ and $\wp'$ to prove

$$\Phi(-z) = \ominus\Phi(z),$$

where $\ominus(x, y) = (x, -y)$ and $\ominus O = O$.

Intent. Write one group element in torus, cubic, and divisor-class notation.

Exercise 9.5. For $z \in T$, write the same group element in the three notations

$$z \in T, \qquad \Phi(z) \in E(\mathbb{C}), \qquad [\Phi(z)] - [O] \in \mathrm{Pic}^0(E).$$

Give the two explicit formulas

$$z \longmapsto [\wp(z) : \wp'(z) : 1], \qquad P \longmapsto [P] - [O]$$

that identify these notations.

Intent. Check the generic injectivity mechanism of the Weierstrass map and mark the theorem boundary.

Exercise 9.6. Assume $\Phi(z) = \Phi(w)$ in the affine chart. Compute the two equalities

$$\wp(z) = \wp(w), \qquad \wp'(z) = \wp'(w).$$

For a generic value of $\wp$, the equation $\wp(t) = \wp(z)$ has the two possibilities $t \equiv z$ and $t \equiv -z$ modulo Λ. Use the oddness of $\wp'$ to show how the second coordinate removes the generic z versus $-z$ ambiguity. State which exceptional half-period and point-at-infinity cases still belong to the full Weierstrass embedding theorem.

Chapter 10

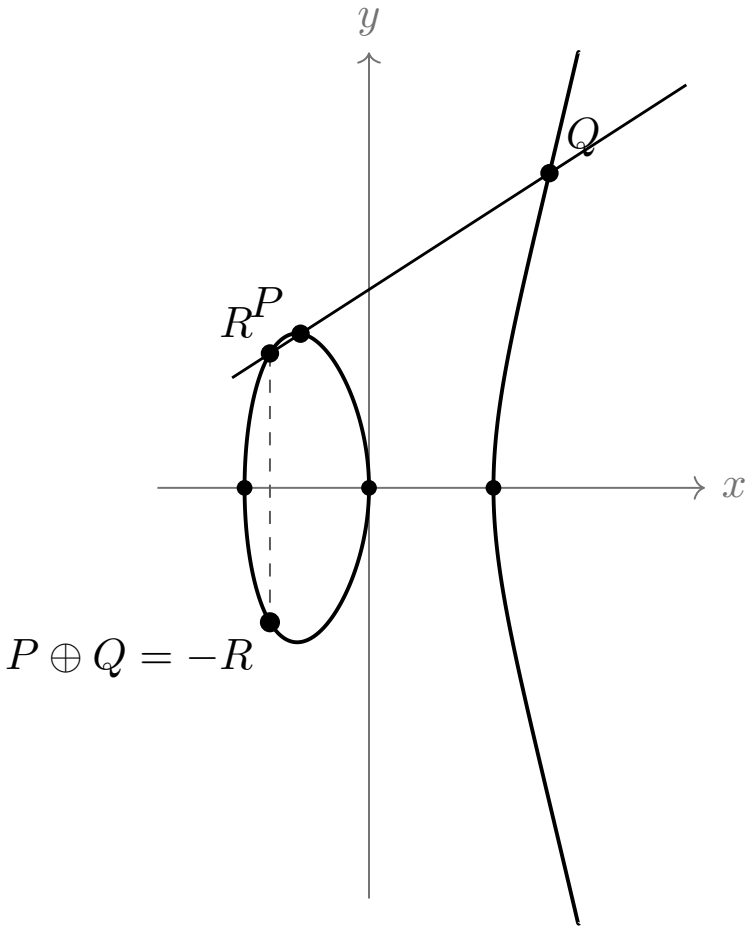
The Plane Cubic and Its Group Law

Let E be the smooth cubic from chapter 9, with identity point $O = [0 : 1 : 0]$. A line in $\mathbb{CP}^2$ meets a cubic in three points counted with multiplicity. This is the only geometric input needed for the chord– tangent rule.

Definition 10.1. Given $P, Q \in E$, draw the line through P and Q, using the tangent line if $P = Q$. Let R be the third intersection point with E. Define

$$P \boxplus Q = \ominus R,$$

where $\ominus(x, y) = (x, -y)$ in the affine chart and $\ominus O = O$.



The reflection step is often the mysterious part. It is not arbitrary: a line through P, Q, R cuts a principal divisor

$$[P] + [Q] + [R] - 3[O].$$

By Abel's theorem, this means $P + Q + R = O$ in the group transported from the torus. Therefore the group sum of P and Q is $-R$.

Proposition 10.2. *If a projective line L meets E in P, Q, R, counted with multiplicity, then $[P] + [Q] + [R] - 3[O]$ is principal.*

Proof sketch. The quotient of the linear form defining L by the linear form defining the line at infinity is a meromorphic function on E. Its zeros are the intersections with L, and its poles are the triple intersection with the line at infinity at O. □

Exercises

Intent. See tangent lines as limiting secants in the nonvertical affine case.

Exercise 10.1. Let P and Q be nearby affine points on a smooth cubic, and assume the tangent at P is not vertical. Let $Q \to P$. Compute the limiting slope of the chord line in local affine coordinates. State why the limiting line has intersection multiplicity 2 at P.

Intent. Derive the chord formula from substitution and Vieta's formula.

Exercise 10.2. For $y^2 = 4x^3 - g_2x - g_3$, derive the formula

$$x_{P \boxplus Q} = \frac{m^2}{4} - x_P - x_Q$$

when the chord has slope m. Do this by substituting $y = mx + b$, comparing the x^2-coefficient of the resulting cubic, and applying Vieta's formula.

Intent. Compute the vertical-line case that gives inverses on the cubic.

Exercise 10.3. If $P = (x, y)$, compute the intersections of the vertical line $X = xZ$ with the cubic. Show that they are P, $\ominus P = (x, -y)$, and O, counted with multiplicity when necessary.

Intent. Compute the line divisor directly from the meromorphic function ℓ/Z.

Exercise 10.4. Let $\ell(X, Y, Z) = 0$ be a projective line meeting E in P, Q, R, counted with multiplicity, and let $h = Z$ define the line at infinity. Compute the divisor of the meromorphic function ℓ/h on E: identify its zeros from $\ell|_E = 0$, identify its poles from $Z|_E = 0$, and use the fact that the line at infinity meets the Weierstrass cubic triply at $O = [0 : 1 : 0]$. Conclude that

$$\operatorname{div}(\ell/h) = [P] + [Q] + [R] - 3[O].$$

Intent. Turn the line divisor into the group equation $P + Q + R = O$ by Abel sums.

Exercise 10.5. Assume Abel's theorem and the transported group law on E. Use the line divisor $[P] + [Q] + [R] - 3[O]$ to compute the Abel sum and prove that $P + Q + R = O$.

Chapter 11

Why Chord–Tangent Equals Torus Addition

We now put the two laws next to each other. The analytic law is

$$\Phi(u) \oplus \Phi(v) = \Phi(u+v).$$

The geometric law is the chord–tangent operation $\boxplus$. The main theorem says these are the same.

Theorem 11.1 (Transport theorem)**.** *For all $P, Q \in E(\mathbb{C})$,*

$$P \oplus Q = P \boxplus Q.$$

Proof. Write $P = \Phi(u)$, $Q = \Phi(v)$, and let R be the third intersection of the line through P and Q. By proposition 10.2, $[P] + [Q] + [R] - 3[O]$ is principal. Abel's theorem therefore gives

$$u + v + r \equiv 0 \pmod{\Lambda},$$

where $R = \Phi(r)$. Hence $R = \Phi(-(u+v))$. Since $\wp$ is even and $\wp'$ is odd, reflecting R gives $\Phi(u+v)$. Thus the chord–tangent result is exactly the transported torus sum. □

Corollary 11.2. *The chord–tangent law is associative and commutative.*

Proof. It is transported from the associative and commutative law on $\mathbb{C}/\Lambda$. □

No arbitrary convention. The reflection is forced by the equation $P+Q+R = O$. The third intersection is not the sum; its inverse is the sum.

Exercises

Intent. Locate the exact Abel-sum calculation inside the transport proof.

Exercise 11.1. In the proof of theorem 11.1, write the line divisor $[P] + [Q] + [R] - 3[O]$, choose torus preimages $u, v, r, 0$, and compute its Abel sum. Identify the exact equality where Abel's theorem is used.

Intent. Compare direct cubic associativity with linear associativity on the torus.

Exercise 11.2. List the third-intersection computations needed to compare

$$(P \boxplus Q) \boxplus R \quad \text{and} \quad P \boxplus (Q \boxplus R)$$

directly on the cubic. Then write the corresponding torus calculation

$$(u + v) + w = u + (v + w)$$

and explain which side is linear.

Intent. Verify the identity element using the projective vertical line $X = xZ$.

Exercise 11.3. For $P = (x, y)$, compute the three intersections of the projective line $X = xZ$ with the cubic. Check that this line contains P and $O = [0 : 1 : 0]$. Use the reflection step to show that $O \boxplus P = P$.

Intent. Derive $P \boxplus Q = \Phi(u + v)$ from the line divisor and Abel sum.

Exercise 11.4. Let $P = \Phi(u)$ and $Q = \Phi(v)$. Give a step-by-step proof that the chord–tangent construction gives $\Phi(u+v)$. Your proof must include the Abel-sum equation $u+v+r \equiv 0 \pmod{\Lambda}$ and the computation $\Phi(-r) = \Phi(u + v)$.

Intent. Identify the reflection step as the torus operation $r \mapsto -r$.

Exercise 11.5. In the previous exercise, compute $r = -(u + v)$. Then use

$$\Phi(-r) = \ominus\Phi(r)$$

to identify the exact point where the reflection $\ominus R$ enters.

Chapter 12

Cohomology, Moduli, and the j-Invariant

The Jacobian also has a cohomological incarnation. The exponential sequence of sheaves

$$0 \to \mathbb{Z} \to \mathcal{O}_E \to \mathcal{O}_E^* \to 0$$

connects integral cycles, holomorphic functions, and line bundles. Taking cohomology yields, in degree one, the complex torus

$$\mathrm{Pic}^0(E) = H^1(E, \mathcal{O}_E)/H^1(E, \mathbb{Z}) \cong \mathbb{C}/\Lambda.$$

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{2\pi i} & \mathcal{O}_E & \xrightarrow{\exp} & \mathcal{O}_E^* & \longrightarrow & 0 \\ & & & & \downarrow & & & & \\ & & & & \mathrm{Pic}^0(E) = \dfrac{H^1(E, \mathcal{O}_E)}{H^1(E, \mathbb{Z})} & \cong & \dfrac{\mathbb{C}}{\Lambda} & & \end{array}$$

This chapter's claim is a structural explanation, not a replacement for the divisor proof. Divisors make the group visible through zeros and poles; sheaf cohomology makes the same group visible through line bundles and periods.

Definition 12.1. The j-invariant of $E : Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3$ is

$$j(E) = 1728\,\frac{g_2^3}{g_2^3 - 27g_3^2}.$$

Scaling a lattice changes g_2 and g_3, but it does not change j. Thus j records the isomorphism class of the complex elliptic curve. Lattice shapes are parametrized by $\tau \in \mathbb{H}$ modulo the modular group $\mathrm{SL}_2(\mathbb{Z})$, and the j-function identifies the quotient $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ with the complex line.

Exercises

Intent. Check that the j-invariant is unchanged by scaling weights.

Exercise 12.1. If $g_2 \mapsto a^{-4}g_2$ and $g_3 \mapsto a^{-6}g_3$, show that j is unchanged. Compute separately the scaling weights of g_2^3, g_3^2, and $\Delta = g_2^3 - 27g_3^2$.

Intent. Recognize the cohomological Jacobian as a complex vector space modulo a lattice.

Exercise 12.2. Use the genus-one facts

$$H^1(E, \mathcal{O}_E) \cong \mathbb{C}, \qquad H^1(E, \mathbb{Z}) \cong \mathbb{Z}^2$$

to write the quotient $H^1(E, \mathcal{O}_E)/H^1(E, \mathbb{Z})$ in the form $\mathbb{C}/\Lambda$. State explicitly what the rank-two lattice Λ is standing for.

Intent. Compute the extra lattice rotations at the special square and hexagonal shapes.

Exercise 12.3. For the lattices $\mathbb{Z} + i\mathbb{Z}$ and $\mathbb{Z} + e^{2\pi i/3}\mathbb{Z}$, compute which rotations preserve the lattice: multiplication by i in the first case and by $e^{2\pi i/3}$ in the second. Record the corresponding rotation orders.

Intent. Separate what cohomology explains from what the chord–tangent proof still needs.

Exercise 12.4. Which part of the three-face theorem is re-expressed by the quotient

$$H^1(E, \mathcal{O}_E)/H^1(E, \mathbb{Z}) \cong \mathbb{C}/\Lambda?$$

Write the map or identification it replaces, and list the two extra calculations still needed for the chord–tangent formula: the line divisor calculation and the Abel-sum calculation.

Chapter 13

Finite Fields and Arithmetic Directions

The theory does not stop over $\mathbb{C}$: it also moves to finite fields and number fields. This is possible because the chord–tangent formulas are rational in the coordinates. Once the cubic equation and the addition formulas are written with coefficients in a field k, the same formulas define a group $E(k)$ whenever the cubic is smooth.

Theorem 13.1 (Hasse bound)**.** *For an elliptic curve $E/\mathbb{F}_p$,*

$$\#E(\mathbb{F}_p) = p + 1 - a_p, \qquad |a_p| \le 2\sqrt{p}.$$

Status. This is a deep theorem, not proved in this textbook. It is included to mark the transition from complex geometry to arithmetic geometry. □

Cryptography uses the finite abelian group $E(\mathbb{F}_p)$. Adding points is fast because it is given by explicit rational formulas. Reversing repeated addition, recovering n from P and nP, is the elliptic-curve discrete logarithm problem.

The Tate module is another arithmetic object. For a prime $\ell \neq p$, the ℓ^n-torsion groups fit into an inverse system, and

$$T_\ell E = \varprojlim E[\ell^n] \cong \mathbb{Z}_\ell^2.$$

If E is defined over $\mathbb{Q}$, the absolute Galois group acts on these torsion points:

$$\rho_\ell : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_2(\mathbb{Z}_\ell).$$

Exercises

Intent. Count finite-field points by a concrete residue table.

Exercise 13.1. Count the points on $y^2 = x^3 + 2x + 1$ over $\mathbb{F}_5$. Make a table for $x = 0, 1, 2, 3, 4$, compute $x^3 + 2x + 1 \pmod 5$, compare with the square set $\{0, 1, 4\}$, and then add the point at infinity.

Intent. Identify the field operations used by the rational addition formulas.

Exercise 13.2. In the chord–tangent formulas, list the field operations that appear $+, -, \times$, and division by a nonzero denominator. From that list, state the calculation rule that lets the same formulas define addition over any field where the cubic is smooth, and name the exceptional denominator cases that must be handled separately.

Intent. Place a finite-field point count inside Hasse's interval.

Exercise 13.3. Write the Hasse bound as the explicit interval

$$p + 1 - 2\sqrt{p} \leq \#E(\mathbb{F}_p) \leq p + 1 + 2\sqrt{p}.$$

Then interpret the interval geometrically.

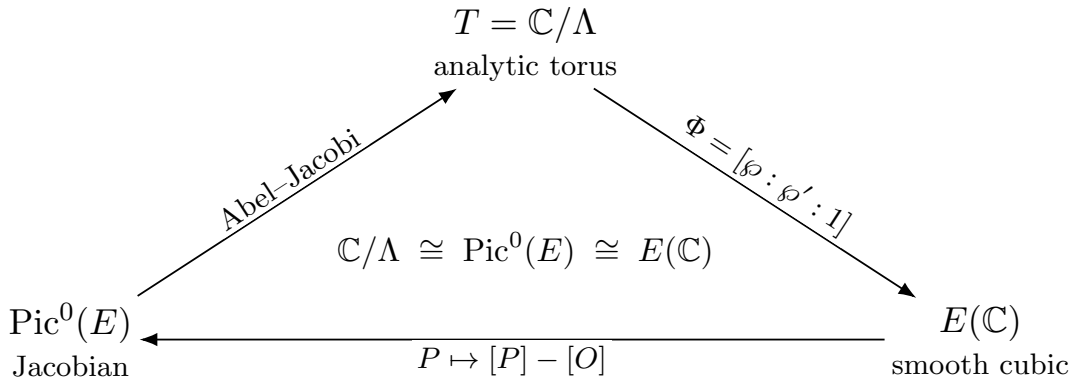
Intent. Separate algebraic ingredients valid over general fields from complex-analytic ones.

Exercise 13.4. Make a two-column table. In the first column, list algebraic ingredients that still make sense over a general field: the cubic equation, O, divisors, and rational chord–tangent formulas. In the second column, list analytic ingredients that depend on $\mathbb{C}$: lattices, $\mathbb{C}/\Lambda$, $\wp$, periods, and complex Lie groups.

Chapter 14

Synthesis: The Three Faces of One Group

We can now assemble the whole argument without skipping any major arrow.



1. A lattice $\Lambda \subset \mathbb{C}$ gives a compact genus-one complex Lie group $T = \mathbb{C}/\Lambda$.
2. Elliptic functions are meromorphic functions on T.
3. The Weierstrass function $\wp$ has one double pole at O, and $\wp'$ has one triple pole.
4. The differential equation $\wp'^2 = 4\wp^3 - g_2\wp - g_3$ produces a cubic E.
5. Riemann–Roch explains why $1, \wp, \wp'$ are exactly the functions needed for the projective map $T \to \mathbb{CP}^2$.
6. The map $\Phi = [\wp : \wp' : 1]$ identifies T with the smooth cubic $E(\mathbb{C})$.
7. Divisors and Abel's theorem identify $\mathrm{Pic}^0(E)$ with the same group.
8. A line meeting E in P, Q, R cuts a principal divisor $[P] + [Q] + [R] - 3[O]$.
9. Abel's theorem gives $P + Q + R = O$, so the group sum of P and Q is the reflection of R.
10. Because the addition formulas are rational, the same cubic opens into arithmetic over $\mathbb{Q}$ and finite fields.

Theorem 14.1 (Three-face theorem)**.** *Over $\mathbb{C}$, the analytic torus $\mathbb{C}/\Lambda$, the Jacobian $\mathrm{Pic}^0(E)$, and the smooth cubic $E(\mathbb{C})$ are isomorphic as complex Lie groups. Equivalently, the identifying maps are biholomorphic group isomorphisms. Under these identifications, the chord–tangent rule on E is the group law on $\mathbb{C}/\Lambda$ transported by $\Phi = [\wp : \wp' : 1]$.*

Proof. The analytic-to-algebraic identification is theorem 9.1. The geometric identification with the Jacobian is corollary 8.2 transported through Φ. The equality between transported addition and chord–tangent addition is theorem 11.1. □

Exercises

Intent. Recover the forced reflection step from the line-divisor Abel equation.

Exercise 14.1. Compute the line-divisor relation for collinear P, Q, R:

$$[P] + [Q] + [R] - 3[O] \in \mathrm{Prin}(E).$$

Apply Abel's theorem to get $P + Q + R = O$, and solve this equation for the chord–tangent sum $P \boxplus Q$.

Intent. Organize the three languages and the maps between them.

Exercise 14.2. Make a three-column table with headings analytic, divisor-theoretic, and algebraic. Place $\mathbb{C}/\Lambda$, $\mathrm{Pic}^0(E)$, $E(\mathbb{C})$, $\wp$, divisors, and lines in the appropriate columns. Add a second row containing the explicit maps

$$\Phi = [\wp : \wp' : 1], \qquad P \mapsto [P] - [O],$$

and the Abel–Jacobi map.

Intent. Map each chapter block to the calculation type it contributes.

Exercise 14.3. Make a chapter-to-method table. For each block of chapters, record one calculation type: residue or integral calculation, lattice linear-algebra calculation, pole-order dimension count, divisor coefficient calculation, line intersection calculation, cohomology quotient calculation, or finite-field point count.

Intent. Audit the two main maps as complex Lie group isomorphisms and mark theorem inputs.

Exercise 14.4. Make a map-check table for

$$\mathbb{C}/\Lambda \cong \mathrm{Pic}^0(E) \cong E(\mathbb{C})$$

as complex Lie groups. Use rows Φ and $P \mapsto [P] - [O]$, and columns: formula, source, target, group-law compatibility, holomorphicity input, and inverse or bijectivity input. In each entry, mark whether it is a direct calculation from the exercises or an accepted named theorem.

Intent. Compute final Jacobian addition from the Abel sum of $[P_1] + [P_2] - [S] - [O]$.

Exercise 14.5. Give a final dictionary for a point $z \in \mathbb{C}/\Lambda$. Include its torus notation, its cubic notation, its divisor-class notation, and the rule for adding two such points in each language. Write the formulas for $\Phi(z)$, $[\Phi(z)] - [O]$, $\Phi(z_1 + z_2)$, and the corresponding sum in $\mathrm{Pic}^0(E)$. Then, for $P_i = \Phi(z_i)$ and $S = \Phi(z_1 + z_2)$, compute the Abel sum of

$$[P_1] + [P_2] - [S] - [O]$$

and use Abel's theorem to prove

$$([P_1] - [O]) + ([P_2] - [O]) = [S] - [O]$$

in $\mathrm{Pic}^0(E)$.

Appendix A

Complex-Analysis Toolkit

This appendix records the analytic facts used in the chapters.

A.1 Residues

If f has Laurent expansion $f(z) = \sum a_n (z-p)^n$, then $\operatorname{Res}_p(f\,dz) = a_{-1}$. The residue theorem says that the integral of a meromorphic 1-form around a contour is $2\pi i$ times the sum of residues inside.

A.2 Argument Principle

For meromorphic f,

$$\frac{1}{2\pi i}\int_\gamma \frac{f'}{f}\,dz$$

counts zeros minus poles inside γ, with multiplicity.

A.3 Compact Residue Theorem

On a compact Riemann surface, the sum of residues of a meromorphic 1-form is zero. This is the global form of the puncture-and-Stokes argument from the midterm.

A.4 Maximum Modulus

A holomorphic function on a compact connected Riemann surface is constant. This is used repeatedly to show that a holomorphic elliptic function must be constant.

Appendix B

Exam-to-Textbook Bridge

B.1 From the Midterm

- Differential forms and vector calculus become the language of 1-forms and residues in chapter 1.
- The winding form is the prototype for a closed local object that is not globally exact.
- Cauchy–Green and Stokes explain why deleting small disks around singularities leaves residue contributions.
- The Euler characteristic computation for the torus supplies $g = 1$, used in chapter 7.

B.2 From the Final

- Compact Riemann surfaces and meromorphic maps are the setting of chapter 2.
- The theorem that principal divisors have degree zero appears as proposition 6.2.
- Riemann–Roch supplies the dimension count $\ell(nO) = n$ in chapter 7.
- The final problem on the Weierstrass equation becomes theorem 5.2.
- Base point freeness and separation are the background for why a three-dimensional linear system can embed a genus-one curve in projective space.

The exam arc. The midterm gives the obstruction philosophy; the final gives the compact Riemann surface technology. Together they lead to the equivalence $\mathbb{C}/\Lambda \cong \mathrm{Pic}^0(E) \cong E(\mathbb{C})$ as complex Lie groups, not merely as abstract abelian groups.

Appendix C

Solutions to Exercises

C.1 Chapter 1

1.1. Write

$$\omega_{\text{wind}} = P\,dx + Q\,dy, \qquad P = \frac{-y}{x^2+y^2}, \quad Q = \frac{x}{x^2+y^2}.$$

Away from the origin,

$$\frac{\partial Q}{\partial x} = \frac{y^2 - x^2}{(x^2+y^2)^2} = \frac{\partial P}{\partial y}.$$

Thus

$$d\omega_{\text{wind}} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx \wedge dy = 0.$$

$$\boxed{\omega_{\text{wind}} = P\,dx + Q\,dy} \rightarrow \boxed{d\omega = (Q_x - P_y)\,dx \wedge dy} \rightarrow \boxed{\begin{matrix} Q_x = P_y \\ d\omega_{\text{wind}} = 0 \end{matrix}}$$

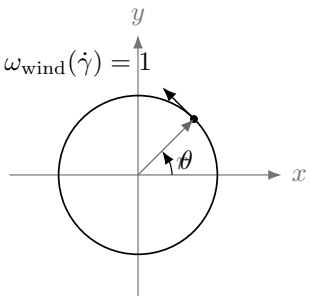
1.2. With $x = \cos t$, $y = \sin t$, one has

$$dx = -\sin t\,dt, \qquad dy = \cos t\,dt.$$

Thus

$$-y\,dx + x\,dy = \sin^2 t\,dt + \cos^2 t\,dt = dt,$$

and the denominator is 1. The integral from 0 to 2π is 2π.

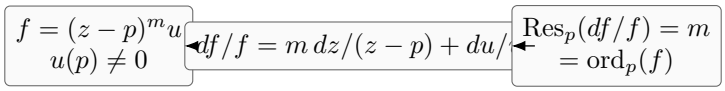


On S^1, the winding form pulls back to $d\theta = dt$.

1.3. Write $f = (z-p)^m u(z)$, where $u(p) \neq 0$. Then

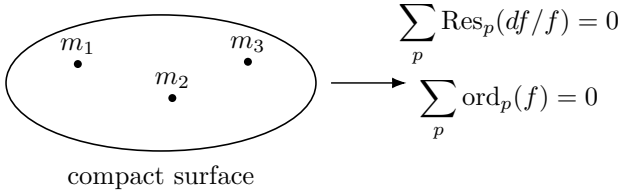
$$\frac{df}{f} = m\,\frac{dz}{z-p} + \frac{du}{u}.$$

The second term is holomorphic near p, so the residue is $m = \operatorname{ord}_p(f)$.



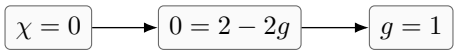
1.4. By Exercise 1.3, the residue of df/f at p is $\operatorname{ord}_p(f)$. On a compact Riemann surface the sum of residues of a meromorphic one-form is zero. Therefore

$$0 = \sum_p \operatorname{Res}_p(df/f) = \sum_p \operatorname{ord}_p(f) = \deg \operatorname{div}(f).$$

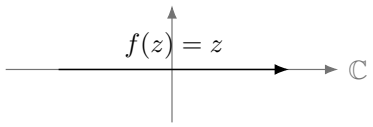


C.2 Chapter 2

2.1. If $\chi = 0$, then $0 = 2 - 2g$, so $g = 1$.

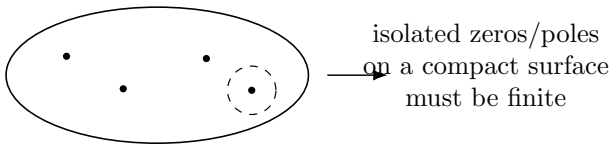


2.2. Noncompact surfaces need not give compact images, so a maximum of $|f|$ may not exist. For example, $f(z) = z$ on $\mathbb{C}$ is holomorphic and nonconstant.



noncompact domain: $|f|$ can grow without a maximum

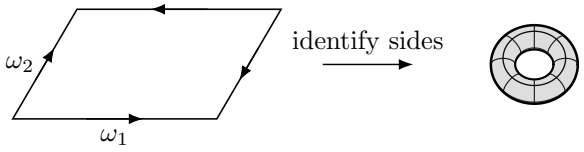
2.3. For a nonzero meromorphic function, zeros and poles are isolated. An infinite subset of a compact surface has an accumulation point. That would contradict isolation: a holomorphic function with accumulating zeros is identically zero locally, and poles are isolated by the local form $(z-p)^{-m}u(z)$. Hence the zero and pole sets are finite.



2.4. A parallelogram with opposite sides identified has one face, two edge classes, and one vertex class. Hence

$$\chi = 1 - 2 + 1 = 0.$$

Since $\chi = 2 - 2g$, the quotient has $g = 1$. Thus the quotient surface is a torus.



one face, two edge classes,
one vertex class

C.3 Chapter 3

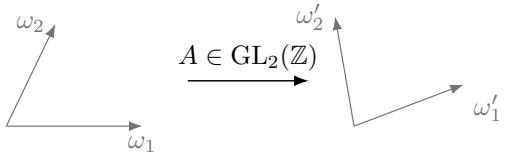
3.1. If $u' = u + \lambda$ and $v' = v + \mu$, then

$$u' + v' = u + v + \lambda + \mu.$$

Since $\lambda + \mu \in \Lambda$, this represents the same class as $u + v$.

$$\boxed{\begin{matrix} u' = u + \lambda \\ v' = v + \mu \end{matrix}} \longrightarrow \boxed{u' + v' = u + v + \lambda + \mu} \longrightarrow \boxed{[u' + v'] = [u + v]}$$

3.2. A matrix in $\mathrm{GL}_2(\mathbb{Z})$ gives integer combinations of ω_1, ω_2, and its inverse also has integer entries. Thus each old basis vector is an integer combination of the new ones and conversely, so the generated subgroup is unchanged.

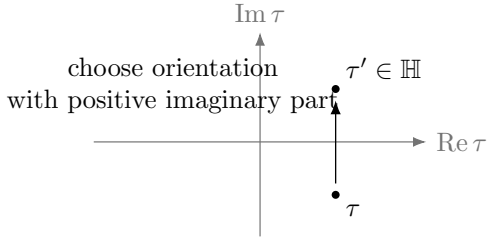


integer change of basis preserves the lattice

3.3. The ratio $\tau = \omega_2/\omega_1$ is non-real because, if $\tau \in \mathbb{R}$, then $\omega_2 = \tau\omega_1$ would make ω_1, ω_2 real-linearly dependent. If $\tau = a + ib$ with $b < 0$, swapping the basis vectors gives the new ratio

$$\frac{\omega_1}{\omega_2} = \frac{1}{\tau} = \frac{a - ib}{a^2 + b^2},$$

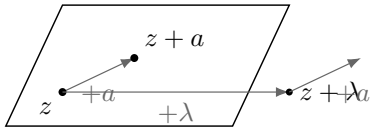
whose imaginary part is $-b/(a^2 + b^2) > 0$. Hence an oriented basis can be chosen with $\tau \in \mathbb{H}$.



3.4. The map is well-defined because if $z' = z + \lambda$, then

$$z' + a = z + a + \lambda,$$

so $[z'+a] = [z+a]$. It is holomorphic because it is induced locally by the holomorphic map $z \mapsto z+a$ on $\mathbb{C}$. Its inverse is translation by $-a$, so it is biholomorphic.

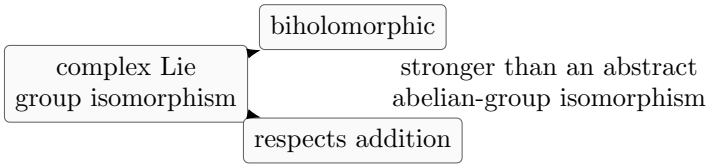


$$[z] = [z + \lambda] \text{ and } [z + a] = [z + \lambda + a].$$

3.5. For a map $F : G \to H$, the checklist is:

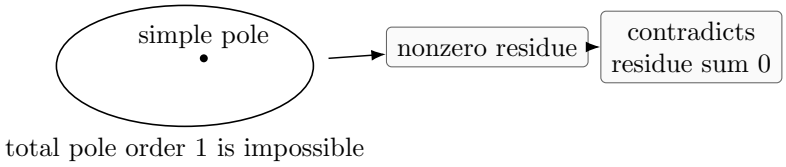
condition	calculation or property
group homomorphism	$F(x + y) = F(x) + F(y)$
holomorphic map	F is holomorphic in local complex charts
holomorphic inverse	F^{-1} exists and is holomorphic.

Thus an isomorphism of complex Lie groups is simultaneously a biholomorphism and a group isomorphism, not merely an abstract abelian-group isomorphism.



C.4 Chapter 4

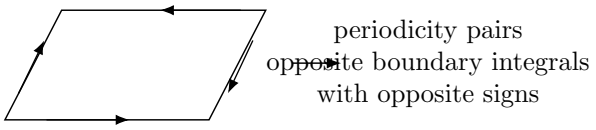
4.1. A nonconstant elliptic function cannot be holomorphic, so it has a pole. If the total pole order were 1, there would be exactly one simple pole. Its Laurent expansion would have a nonzero coefficient of $(z - a)^{-1}$, hence a nonzero residue. This contradicts the residue-sum law for elliptic functions. Therefore the total pole order is at least 2.



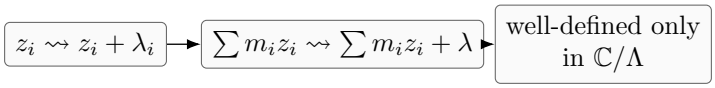
4.2. Choose a period parallelogram whose boundary contains no zeros or poles of f. Apply the argument principle to f on its boundary:

$$\frac{1}{2\pi i} \int_{\partial P} \frac{f'}{f}\, dz.$$

Opposite sides cancel by periodicity. The integral is therefore zero, and it equals the number of zeros minus the number of poles, counted with multiplicity.



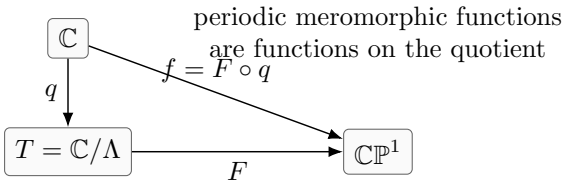
4.3. Changing a lift z_i by a lattice element changes the weighted sum by an element of Λ. Therefore the Abel sum is intrinsic only modulo Λ, that is, as a point of $\mathbb{C}/\Lambda$.



4.4. If F is meromorphic on T, then $F \circ q$ is meromorphic on $\mathbb{C}$ and satisfies

$$F(q(z+\lambda)) = F(q(z))$$

for all $\lambda \in \Lambda$. Conversely, if f is elliptic, define $F([z]) = f(z)$. Periodicity makes this well-defined, and the quotient charts show that F is meromorphic on T.



4.5. For

$$\operatorname{div}(f) = [a_1] + [a_2] - [b_1] - [b_2],$$

the third Liouville law gives

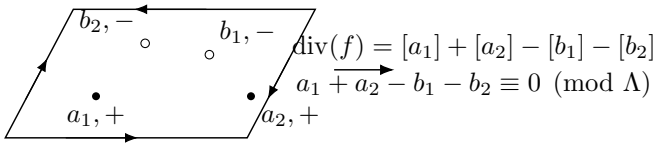
$$a_1 + a_2 - b_1 - b_2 \in \Lambda.$$

Modulo Λ, this is

$$a_1 + a_2 - b_1 - b_2 \equiv 0 \pmod{\Lambda}.$$

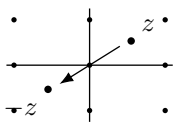
By the definition of the Abel-sum map, this says

$$\mathfrak{s}(\operatorname{div}(f)) = 0 \in T.$$



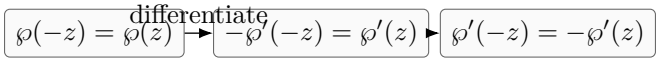
C.5 Chapter 5

5.1. Replacing z by $-z$ permutes the lattice terms $\lambda \mapsto -\lambda$ and leaves the series unchanged. Hence $\wp(-z) = \wp(z)$.



the lattice is unchanged by
$\lambda \mapsto -\lambda$, so $\wp$ is even

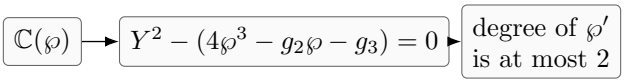
5.2. The derivative of an even function is odd: differentiating $\wp(-z) = \wp(z)$ gives $-\wp'(-z) = \wp'(z)$, so $\wp'(-z) = -\wp'(z)$.



5.3. The equation

$$\wp'^2 = 4\wp^3 - g_2\wp - g_3$$

is a polynomial equation for $\wp'$ with coefficients in $\mathbb{C}(\wp)$. It is quadratic in $\wp'$, so $\wp'$ has degree at most 2 over $\mathbb{C}(\wp)$.



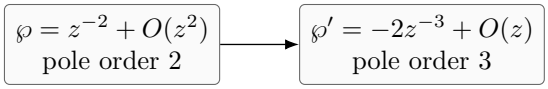
5.4. From the Laurent expansion

$$\wp(z) = z^{-2} + O(z^2),$$

$\wp$ has a pole of order 2 at O. From

$$\wp'(z) = -2z^{-3} + O(z),$$

$\wp'$ has a pole of order 3 at O.

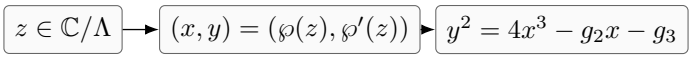


the leading negative power gives the pole order at O

5.5. Substituting $x = \wp(z)$ and $y = \wp'(z)$ into the differential equation gives

$$y^2 = 4x^3 - g_2x - g_3.$$

Thus every point $(\wp(z), \wp'(z))$, away from the pole, lies on the affine cubic.



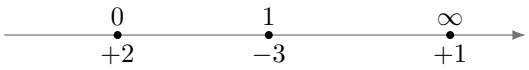
C.6 Chapter 6

6.1. The function $z^2(z-1)^{-3}$ has a zero of order 2 at 0, a pole of order 3 at 1, and a pole/zero contribution at ∞ chosen so the total degree is zero. In the coordinate $w = 1/z$, the function is

$$\frac{w}{(1-w)^3},$$

so it has a zero of order 1 at ∞. Hence

$$\operatorname{div}(z^2(z-1)^{-3}) = 2[0] - 3[1] + [\infty].$$

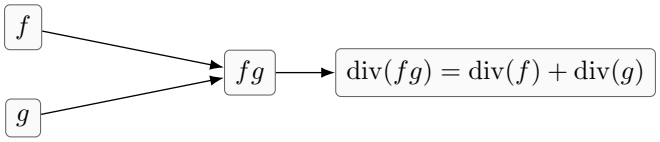


zeros and poles balance to total degree 0

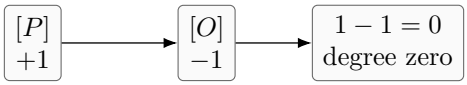
6.2. Products of meromorphic functions add divisors:

$$\operatorname{div}(fg) = \operatorname{div}(f) + \operatorname{div}(g).$$

Inverses negate divisors, and constants give the zero divisor. Since principal divisors have degree zero, $\operatorname{Prin}(X)$ is a subgroup of $\operatorname{Div}^0(X)$.



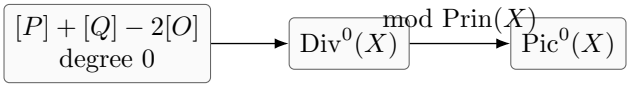
6.3. The coefficients of $[P] - [O]$ are 1 and -1, so the degree is $1 - 1 = 0$.



6.4. The divisor

$$[P] + [Q] - 2[O]$$

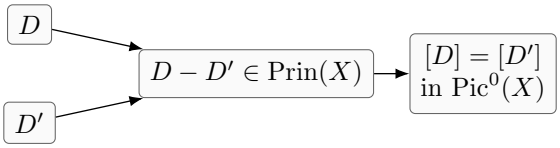
has degree $1 + 1 - 2 = 0$. Therefore it lies in $\operatorname{Div}^0(X)$, and its class modulo principal divisors is an element of $\operatorname{Pic}^0(X)$.



6.5. Two degree-zero divisors D, D' define the same element of $\operatorname{Pic}^0(X) = \operatorname{Div}^0(X)/\operatorname{Prin}(X)$ exactly when

$$D - D' \in \operatorname{Prin}(X).$$

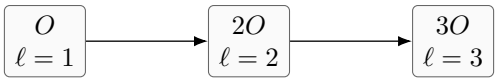
This is the defining equivalence relation in the quotient group.



C.7 Chapter 7

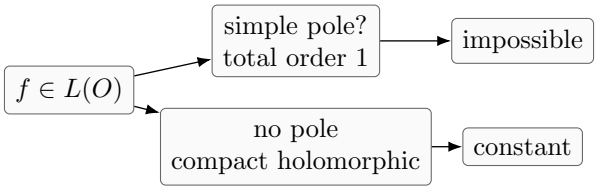
7.1. The dimensions are

$$\ell(O) = 1, \qquad \ell(2O) = 2, \qquad \ell(3O) = 3.$$

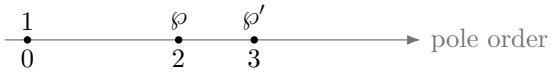


genus-one Riemann–Roch: $\ell(nO) = n$

7.2. A function in $L(O)$ has at most a simple pole at O and no other poles. If it has no pole, it is holomorphic on the compact torus and is constant. If it has a simple pole at O, it has total pole order 1, which is impossible by Exercise 4.1. Hence every element of $L(O)$ is constant.



7.3. The pole orders at O are $0, 2, 3$ for $1, \wp, \wp'$. In a linear relation, the term with the largest pole order cannot be cancelled by terms with smaller pole order. Descending through the possible largest pole orders forces all coefficients to be zero.



the largest pole order in a relation cannot be cancelled

7.4. The constant function satisfies

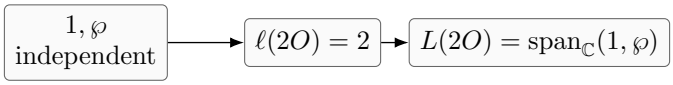
$$\operatorname{div}(1) + 2[O] = 2[O] \geq 0,$$

so $1 \in L(2O)$. Since $\wp$ has no poles away from O and has pole order 2 at O,

$$\operatorname{div}(\wp) + 2[O] \geq 0,$$

so $\wp \in L(2O)$. The pole orders 0 and 2 make $1, \wp$ linearly independent. Since $\ell(2O) = 2$, they form a basis:

$$L(2O) = \operatorname{span}_{\mathbb{C}}(1, \wp).$$

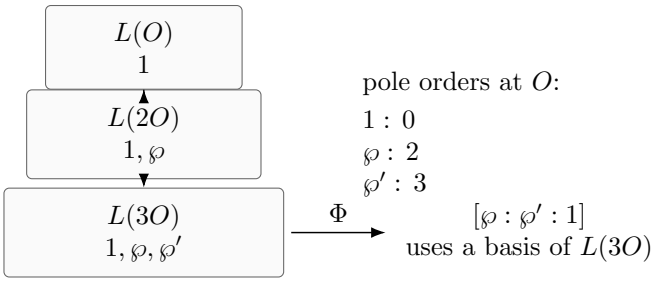


7.5. Membership is checked by the divisor inequalities

$$\operatorname{div}(1) + 3[O] = 3[O] \geq 0, \qquad \operatorname{div}(\wp) + 3[O] \geq 0, \qquad \operatorname{div}(\wp') + 3[O] \geq 0.$$

The second inequality holds because $\wp$ has pole order $2 \leq 3$ at O; the third holds because $\wp'$ has pole order exactly 3 at O. The functions $1, \wp, \wp'$ are linearly independent by pole orders $0, 2, 3$. Since $\ell(3O) = 3$, they form a basis:

$$L(3O) = \operatorname{span}_{\mathbb{C}}(1, \wp, \wp').$$

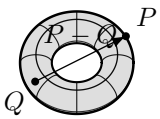


C.8 Chapter 8

8.1. Choose lifts $p, q \in \mathbb{C}$ of P, Q. Then

$$\mathfrak{s}([P] - [Q]) = p - q \pmod{\Lambda}.$$

This is exactly the class $P - Q$ in the quotient group $T = \mathbb{C}/\Lambda$.

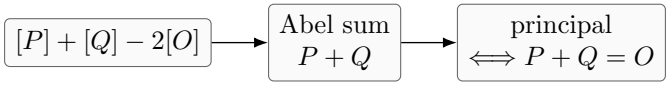


$[P] - [Q]$ has Abel sum $P - Q$ in T

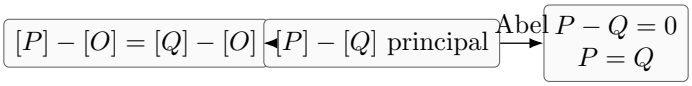
8.2. Choose lifts $p, q, 0 \in \mathbb{C}$ of P, Q, O. The Abel sum is

$$p + q - 2 \cdot 0 \equiv p + q \pmod{\Lambda}.$$

By Abel's theorem, $[P] + [Q] - 2[O]$ is principal exactly when this class is zero, namely when $p + q \equiv 0 \pmod{\Lambda}$, or $P + Q = O$ in T.



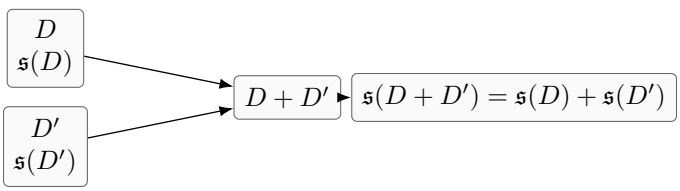
8.3. If $[P] - [O] = [Q] - [O]$, then $[P] - [Q]$ is principal. Abel's theorem gives $P - Q = 0$ in T, hence $P = Q$.



8.4. Write $D = \sum_i n_i[p_i]$ and $D' = \sum_j m_j[q_j]$. Then

$$\mathfrak{s}(D + D') = \sum_i n_i p_i + \sum_j m_j q_j = \mathfrak{s}(D) + \mathfrak{s}(D')$$

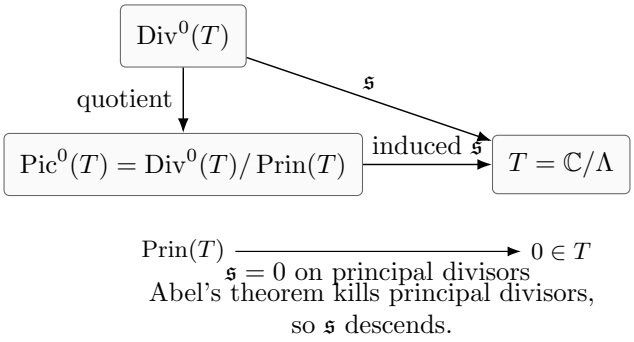
in $\mathbb{C}/\Lambda$.



8.5. If D is replaced by $D + \operatorname{div}(f)$, then

$$\mathfrak{s}(D + \operatorname{div}(f)) - \mathfrak{s}(D) = \mathfrak{s}(\operatorname{div}(f)).$$

The divisor $\operatorname{div}(f)$ is principal, so Abel's theorem gives $\mathfrak{s}(\operatorname{div}(f)) = 0$. Therefore $\mathfrak{s}$ is well-defined on divisor classes in $\operatorname{Pic}^0(T)$. Exercise 8.4 gives additivity, so the induced map is a group homomorphism.

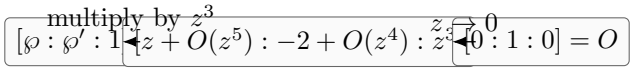


C.9 Chapter 9

9.1. Homogeneous coordinates may be multiplied by z^3:

$$[\wp : \wp' : 1] = [z + O(z^5) : -2 + O(z^4) : z^3],$$

which tends to $[0 : 1 : 0]$.



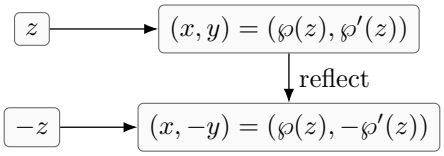
9.2. With $x(z) = \wp(z)$ and $y(z) = \wp'(z)$,

$$x(-z) = \wp(-z) = \wp(z) = x(z),$$

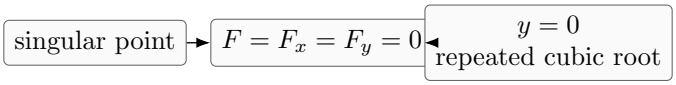
while

$$y(-z) = \wp'(-z) = -\wp'(z) = -y(z).$$

Thus $z \mapsto -z$ fixes the x-coordinate and changes the sign of the y-coordinate.



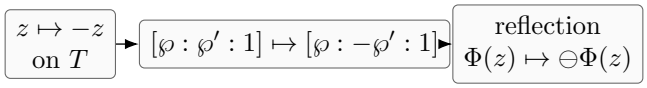
9.3. For $F(x, y) = y^2 - (4x^3 - g_2x - g_3)$, a singular affine point must satisfy $F = 0$, $F_y = 2y = 0$, and $F_x = -12x^2 + g_2 = 0$. Thus $y = 0$ and x is a repeated root of $4x^3 - g_2x - g_3$. Conversely, a repeated root gives such a singular point.



9.4. Since $\wp(-z) = \wp(z)$ and $\wp'(-z) = -\wp'(z)$,

$$\Phi(-z) = [\wp(z) : -\wp'(z) : 1] = \ominus\Phi(z).$$

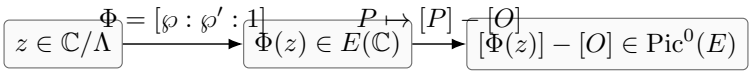
At O, both sides are $O = [0 : 1 : 0]$.



9.5. The torus notation is $z \in T$. The cubic notation is $\Phi(z) \in E(\mathbb{C})$. The divisor-class notation is

$$[\Phi(z)] - [O] \in \mathrm{Pic}^0(E).$$

The first identification is the Weierstrass map Φ, and the second is the Abel-Jacobi divisor-class map $P \mapsto [P] - [O]$.



one group element written in three languages

9.6. If $\Phi(z) = \Phi(w)$ in the affine chart, then equality of projective coordinates gives

$$\wp(z) = \wp(w), \qquad \wp'(z) = \wp'(w).$$

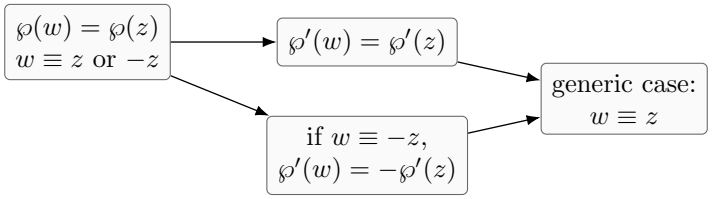
For a generic x-value, the equation $\wp(t) = \wp(z)$ has the two solutions

$$t \equiv z, \qquad t \equiv -z \pmod{\Lambda}.$$

If $w \equiv -z$, then oddness gives

$$\wp'(w) = \wp'(-z) = -\wp'(z).$$

Together with $\wp'(w) = \wp'(z)$, this forces $\wp'(z) = 0$. Thus away from the branch points where $\wp'(z) = 0$, the second coordinate removes the generic z versus $-z$ ambiguity and gives $w \equiv z$. The exceptional half-period cases, where $z \equiv -z$ modulo Λ, and the point O at infinity are handled by the full Weierstrass embedding theorem. This calculation is an injectivity check in the generic case, not a complete proof of the theorem.



exceptional half-periods: $\wp'(z) = 0$

C.10 Chapter 10

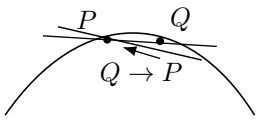
10.1. In affine coordinates, write $P = (x_0, y_0)$ and $Q = (x_0 + h, y(x_0 + h))$. When the tangent is not vertical, the chord slope is

$$\frac{y(x_0 + h) - y(x_0)}{h} \longrightarrow y'(x_0).$$

For an implicit equation $F(x, y) = 0$, this derivative is

$$y'(x_0) = -\frac{F_x(P)}{F_y(P)}$$

when $F_y(P) \neq 0$. The limiting line is the tangent line. Algebraically, its intersection with the cubic has a double zero at P, so P is counted with intersection multiplicity 2.



the chord limit is a tangent and counts P twice

10.2. Substitute $y = mx + b$ into

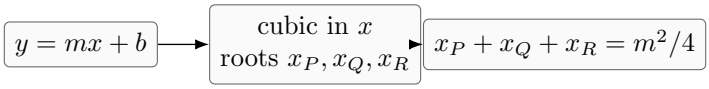
$$y^2 = 4x^3 - g_2x - g_3.$$

The resulting cubic in x has roots x_P, x_Q, x_R, counted with multiplicity in the tangent case. The coefficient of x^2 gives

$$x_P + x_Q + x_R = \frac{m^2}{4}.$$

The chord-tangent sum is the reflection of R, so its x-coordinate is

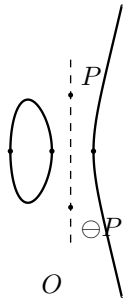
$$x_{P \boxplus Q} = x_R = \frac{m^2}{4} - x_P - x_Q.$$



substitution, then Vieta's formula

10.3. If $P = (x, y)$, the vertical line $X = xZ$ gives the affine intersections (x, y) and $(x, -y)$, since the equation contains y^2. In projective coordinates the same vertical line also meets the cubic at $O = [0 : 1 : 0]$. Thus a vertical line meets P, $\ominus P$, and O, and the third intersection construction gives

$$P \boxplus (\ominus P) = O.$$



a vertical line gives the inverse relation $P \boxplus (\ominus P) = O$

10.4. Let $f = \ell/h = \ell/Z$ on E. The zeros of f are the zeros of $\ell|_E$, namely the three intersections of the line $\ell = 0$ with the cubic:

$$[P] + [Q] + [R],$$

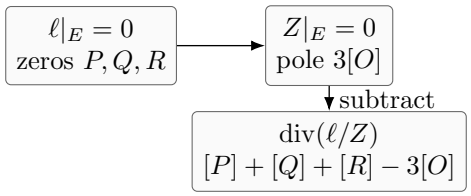
counted with multiplicity. The poles of f are the zeros of $Z|_E$. On the Weierstrass cubic,

$$Z = 0 \quad \Longrightarrow \quad 0 = 4X^3,$$

so the line at infinity meets E only at $O = [0 : 1 : 0]$, with total intersection multiplicity 3. Hence the pole divisor is $3[O]$, and

$$\operatorname{div}(\ell/Z) = [P] + [Q] + [R] - 3[O].$$

Its degree is $1 + 1 + 1 - 3 = 0$.

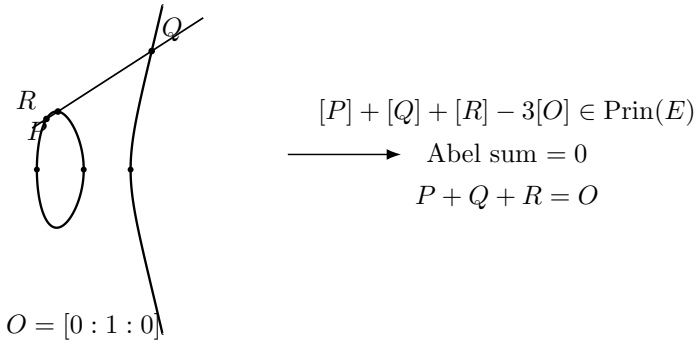


10.5. Since $[P] + [Q] + [R] - 3[O]$ is principal, Abel's theorem says its Abel sum is the identity. Choose torus preimages $p, q, r, 0$ of P, Q, R, O. The Abel sum is

$$p + q + r - 3 \cdot 0 \equiv 0 \pmod{\Lambda}.$$

In the transported group law this is exactly

$$P + Q + R = O.$$



C.11 Chapter 11

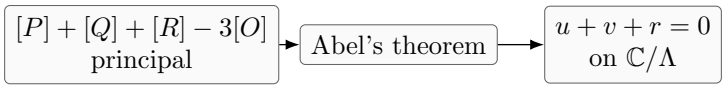
11.1. Abel's theorem is used after the line divisor is known to be principal. After choosing torus preimages u, v, r of P, Q, R, the Abel sum of the divisor is

$$u + v + r - 3 \cdot 0 \equiv u + v + r \pmod{\Lambda}.$$

Abel's theorem converts

$$[P] + [Q] + [R] - 3[O] \in \operatorname{Prin}(E)$$

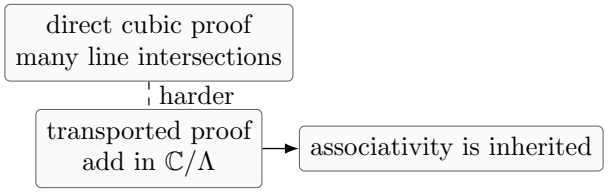
into the group equation $u + v + r = 0$ on the torus.



11.2. A direct proof must compute the third intersection of the line through P, Q, reflect it to get $P \boxplus Q$, then compute the third intersection of the line through $P \boxplus Q, R$. The other bracketing requires the analogous line through Q, R and then the line through $P, Q \boxplus R$. Comparing the final reflected points is a nonlinear intersection calculation. On the torus, the corresponding check is the linear calculation

$$(u+v)+w = u+(v+w).$$

The transported proof avoids the direct intersection comparison by inheriting associativity from $\mathbb{C}/\Lambda$.



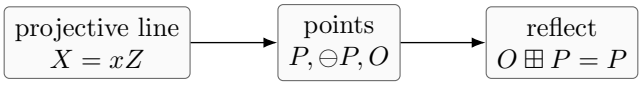
11.3. For $P = (x, y)$, the projective line $X = xZ$ contains

$$P = (x, y, 1), \qquad \ominus P = (x, -y, 1), \qquad O = [0 : 1 : 0].$$

Indeed, O satisfies $X = xZ$ because both sides are 0, and the affine intersections have the same x-coordinate with y-coordinates y and $-y$. Thus the third intersection of the line through O and P is $\ominus P$. Reflecting $\ominus P$ gives P, so

$$O \boxplus P = P.$$

The case $P = O$ gives $O \boxplus O = O$ by the tangent construction at the identity.



11.4. Let $R = \Phi(r)$ be the third intersection of the line through $P = \Phi(u)$ and $Q = \Phi(v)$. The line divisor is principal:

$$[P]+[Q]+[R]-3[O].$$

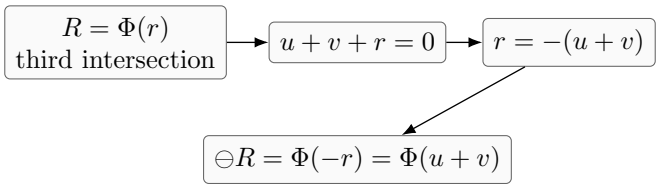
Abel's theorem gives $u + v + r = 0$ in $\mathbb{C}/\Lambda$, hence

$$r = -(u+v).$$

Reflecting $R = \Phi(r)$ gives

$$\ominus R = \Phi(-r) = \Phi(u+v).$$

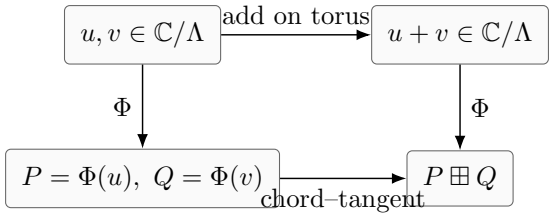
Thus the chord-tangent construction gives the transported torus sum.



11.5. The reflection enters after the third intersection is identified as $R = \Phi(r)$ with $r = -(u+v)$. Reflection on the cubic corresponds to $z \mapsto -z$ on the torus because

$$\Phi(-z) = \ominus\Phi(z).$$

Therefore the reflected point is $\Phi(-r) = \Phi(u+v)$.



The square commutes: $P \boxplus Q = \Phi(u+v)$.

C.12 Chapter 12

12.1. Under $g_2 \mapsto a^{-4}g_2$ and $g_3 \mapsto a^{-6}g_3$, one has

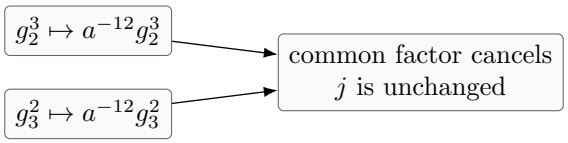
$$g_2^3 \mapsto a^{-12}g_2^3, \qquad g_3^2 \mapsto a^{-12}g_3^2.$$

Therefore

$$\Delta = g_2^3 - 27g_3^2 \mapsto a^{-12}\Delta.$$

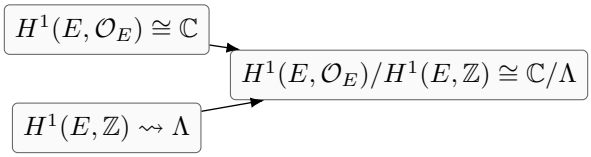
The common factor a^{-12} cancels in

$$j = 1728\frac{g_2^3}{g_2^3 - 27g_3^2}.$$



12.2. In genus one, $H^1(E, \mathcal{O}_E) \cong \mathbb{C}$ as a complex vector space, and $H^1(E, \mathbb{Z}) \cong \mathbb{Z}^2$. Under the natural map into $H^1(E, \mathcal{O}_E)$, the integral classes give a rank-two lattice $\Lambda \subset \mathbb{C}$, generated by two periods. Therefore

$$H^1(E, \mathcal{O}_E)/H^1(E, \mathbb{Z}) \cong \mathbb{C}/\Lambda.$$



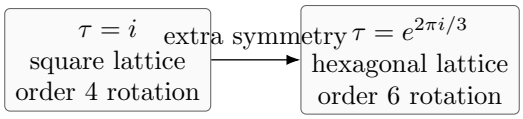
12.3. For $\Lambda = \mathbb{Z} + i\mathbb{Z}$, multiplication by i sends

$$1 \mapsto i, \qquad i \mapsto -1,$$

so it preserves the lattice and has order 4. For $\Lambda = \mathbb{Z} + \tau\mathbb{Z}$, $\tau = e^{2\pi i/3}$, multiplication by τ sends

$$1 \mapsto \tau, \qquad \tau \mapsto \tau^2 = -1 - \tau,$$

so it preserves the lattice and has order 3. Together with multiplication by -1, the hexagonal lattice has rotations of order 6. These rotations suggest extra automorphisms.



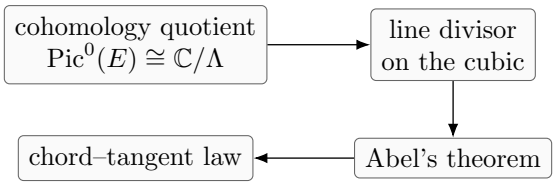
12.4. The quotient re-expresses the identification

$$\mathrm{Pic}^0(E) \cong \mathbb{C}/\Lambda.$$

It explains the Jacobian as a complex torus. It does not by itself prove the chord-tangent formula. Two further calculations are needed: first, the line section calculation

$$\mathrm{div}(L|_E) = [P] + [Q] + [R] - 3[O],$$

and second, the Abel-sum calculation $u + v + r \equiv 0 \pmod{\Lambda}$, which turns that divisor into the group equation.

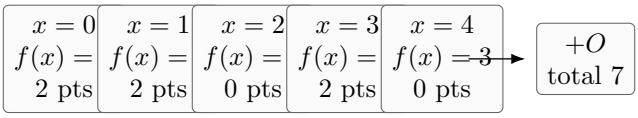


C.13 Chapter 13

13.1. Over $\mathbb{F}_5$, compute $x^3 + 2x + 1$:

$$1, 4, 3, 4, 3$$

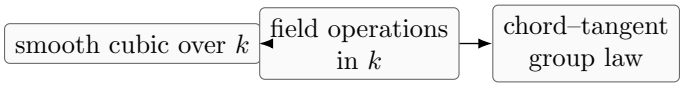
for $x = 0, 1, 2, 3, 4$. The quadratic residues modulo 5 are $0, 1, 4$. Thus $x = 0$ gives two points, $x = 1$ gives two points, $x = 2$ gives none, $x = 3$ gives two points, and $x = 4$ gives none. Adding the point at infinity gives $\#E(\mathbb{F}_5) = 7$.



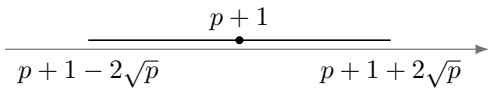
13.2. The chord-tangent formulas use addition, subtraction, multiplication, and division by a nonzero denominator. For example, the slope is

$$m = \frac{y_Q - y_P}{x_Q - x_P}$$

for a nonvertical chord, and the tangent slope is obtained by a similar quotient involving the derivative. Therefore the formulas make sense over any field where the curve is smooth. The exceptional denominator cases are vertical lines, the point at infinity, coincident points requiring the tangent formula, and tangent denominators equal to zero.



13.3. The Hasse bound says that $\#E(\mathbb{F}_p)$ lies within $2\sqrt{p}$ of $p + 1$. Geometrically, the number of finite-field points is close to the projective-line count $p + 1$, with a square-root-size error.



$\#E(\mathbb{F}_p)$ lies in this interval

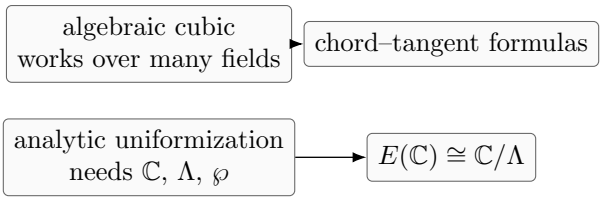
13.4. The requested separation is:

algebraic over a general field	analytic over $\mathbb{C}$
smooth cubic equation	lattice $\Lambda \subset \mathbb{C}$
$O = [0:1:0]$	$\mathbb{C}/\Lambda$
divisors and divisor classes	$\wp, \wp'$
rational chord–tangent formulas	period integrals
$E(k)$ for a field k	complex Lie group structure

The analytic uniformization

$$E(\mathbb{C}) \cong \mathbb{C}/\Lambda$$

uses complex analysis, lattices in $\mathbb{C}$, and meromorphic functions such as $\wp$; that part is special to the complex setting.



C.14 Chapter 14

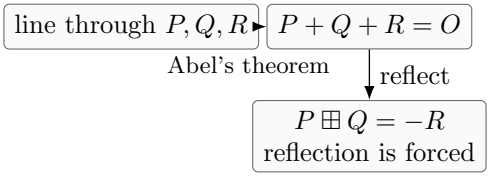
14.1. A line through P, Q, R makes

$$[P] + [Q] + [R] - 3[O]$$

principal. Abel's theorem gives the group equation

$$P + Q + R = O.$$

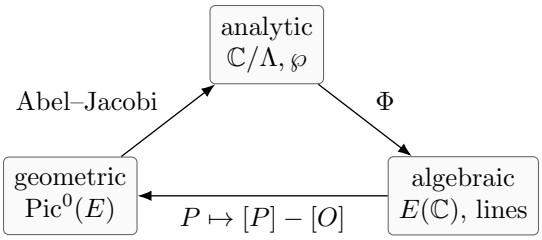
Solving for the group sum gives $P + Q = -R = \ominus R$. The chord–tangent rule therefore defines $P \boxplus Q = \ominus R$; the reflection step is forced by the group equation.



14.2. One possible table is:

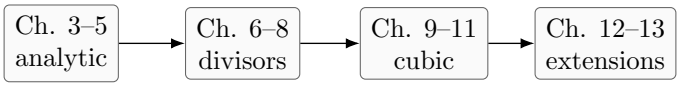
analytic	divisor-theoretic	algebraic
$\mathbb{C}/\Lambda, \wp$	$\mathrm{Pic}^0(E)$, divisors	$E(\mathbb{C})$, lines
$\Phi = [\wp : \wp' : 1]$	Abel–Jacobi	$P \mapsto [P] - [O]$

The first map sends the torus to the cubic, and the second sends a cubic point to its degree-zero divisor class.



14.3. A compact chapter-to-method table is:

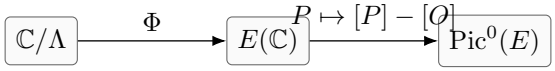
chapters	viewpoint	calculation type
1–2	complex-analysis background	residues, integrals, genus
3–5	analytic torus	lattice algebra, pole orders
6–8	divisors and Jacobian	divisor coefficients, Abel sums
9–11	plane cubic	Vieta and line intersections
12	cohomology and moduli	quotients and scaling weights
13	arithmetic	finite-field point counts



14.4. A compact map-check table is:

map	formula	source	target	group-law compatibility	holomorphicity input	inverse or bijectivity input
Φ	$z \mapsto [\wp(z) : \wp'(z) : 1]$	$\mathbb{C}/\Lambda$	$E(\mathbb{C})$	direct transport calculation: $\Phi(u+v) = \Phi(u) \boxplus \Phi(v)$ after Ex. 11.4	calculation with meromorphic coordinates and the projective limit at O; global extension is theorem input	Weierstrass embedding theorem; Ex. 9.6 gives only the generic injectivity calculation
AJ	$P \mapsto [P] - [O]$	$E(\mathbb{C})$	$\mathrm{Pic}^0(E)$	Abel-sum calculation in Ex. 14.5 plus Abel's theorem	accepted Jacobian/Abel-Jacobi theorem input	Abel's theorem and genus-one representative statement

Thus the final isomorphism as complex Lie groups uses direct calculations for the formulas, Abel sums, reflection step, and transported addition, and uses named theorems for the global biholomorphic and Jacobian-identification inputs.



both maps preserve complex structure and addition

14.5. A point $z \in \mathbb{C}/\Lambda$ is written on the cubic as

$$\Phi(z) = [\wp(z) : \wp'(z) : 1],$$

and in the Jacobian as

$$[\Phi(z)] - [O].$$

Adding z_1, z_2 on the torus gives $z_1 + z_2$. On the cubic this is

$$\Phi(z_1) \oplus \Phi(z_2) = \Phi(z_1 + z_2),$$

which equals the chord-tangent construction. In the Jacobian it is addition of divisor classes:

$$([\Phi(z_1)] - [O]) + ([\Phi(z_2)] - [O]).$$

If $S = \Phi(z_1 + z_2)$, then Abel's theorem gives

$$[\Phi(z_1)] + [\Phi(z_2)] - [S] - [O] \in \mathrm{Prin}(E)$$

because its Abel sum is

$$z_1 + z_2 - (z_1 + z_2) - 0 \equiv 0 \pmod{\Lambda}.$$

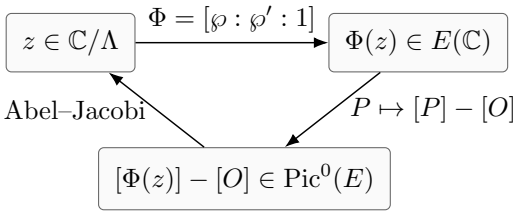
Therefore the class of this divisor is zero in $\mathrm{Pic}^0(E)$, so

$$[\Phi(z_1)] + [\Phi(z_2)] - 2[O] = [S] - [O].$$

Equivalently,

$$([P_1] - [O]) + ([P_2] - [O]) = [S] - [O]$$

for $P_i = \Phi(z_i)$ and $S = \Phi(z_1 + z_2)$.



one group element, three notations

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